

THREAT HUNTING & DIGITAL FORENSICS

Windows Endpoint Investigation Case Study & Evidence Record

Completed Investigation Workbook • Portfolio Edition

Analyst	James Ilemona Akor
Case / Lab ID	Lab-001 (Portfolio ref: M10-DFIR-001)
Investigation period	1–2 August 2026
Environment	Isolated Windows 11 virtual machine
Status	Completed laboratory investigation; evidence limitations documented

Portfolio positioning

This document presents the work as an end-to-end investigation rather than a collection of tool demonstrations. It records what was observed, what was verified, what could not be confirmed, and how the evidence was handled.

Controlled laboratory exercise • Threat Hunting & Digital Forensics

Document Control

Field	Recorded value
Analyst	James Ilemona Akor
Case / Lab ID	Lab-001 (Portfolio ref: M10-DFIR-001)
Environment	Isolated Windows 11 virtual machine under Oracle VirtualBox
Primary host identity observed	James_Akor / James_
Primary evidence	WKSTN11_RAMcapture_1AUG2026.raw; WKSTN11_DiskImage_1AUG2026.E01
Primary tools	Event Viewer; Magnet RAM Capture 1.2.0; FTK Imager 4.7.1.2; Volatility 3 Framework 2.28.0; Autopsy 4.23.1; QuickStego; ExifTool 13.59
Investigation status	Completed laboratory investigation with evidence limitations documented

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1. Executive Summary³

I conducted a controlled digital forensics and threat-hunting investigation against an isolated Windows 11 virtual machine. The exercise demonstrated a defensible workflow from suspicious account activity through evidence preservation, memory analysis, disk examination, hidden-content analysis and final reporting.

The strongest confirmed finding was the creation of a local account named `hacker_backdoor` followed approximately one minute later by its addition to the local Administrators group. Windows Security Event ID 4720 records account creation at 10:39:47 AM on 1 August 2026, while Event ID 4732 records membership in `BUILTIN\Administrators` at 10:40:41 AM.

The retained Event ID 4624 evidence requires an important qualification: the selected event shows Logon Type 5 (service logon), not Logon Type 10 (Remote Desktop). I therefore do not claim that the retained screenshot proves an RDP logon. Volatility independently showed TCP/3389 listening, which establishes RDP availability but not a completed RDP session.

Evidence preservation was demonstrated by capturing volatile memory with Magnet RAM Capture, transferring the resulting RAW file, and creating a segmented E01 disk image with FTK Imager. FTK verification reported matching verification records and no bad blocks. Volatility successfully parsed the memory image and identified the capture utility, OpenSSH, Explorer and listening services. Autopsy identified the main Windows partition, a large deleted-file result set dominated by system/application artefacts, and a BAM artefact confirming recent execution of `MRCv120.exe`.

The hidden-content exercise recovered the synthetic message “A fake set of credentials” from the QuickStego test image. ExifTool then demonstrated a clear format and size difference between the JPEG and BMP versions while retaining the same 1280×960 dimensions.

Analyst assessment

In a real enterprise, the creation of an unexpected local account followed by immediate membership in the local Administrators group would justify escalation and containment review. The evidence supports the account-creation and privilege-assignment findings; it does not independently prove that `hacker_backdoor` completed an RDP logon.

Confirmed Findings at a Glance

ID	Evidence	Assessment	Disposition
F-001	Event 4720 — <code>hacker_backdoor</code> created at 10:39:47	Confirmed account-management event	Escalate / validate
F-002	Event 4732 — <code>hacker_backdoor</code> added to <code>BUILTIN\Administrators</code> at 10:40:41	Confirmed privilege assignment; ~54 seconds after creation	Escalate / contain if unauthorized
F-003	Event 4624 retained evidence — Logon Type 5	Does not prove RDP	Correlate with Type 10 / 4778 / 4779
F-004	Volatility netscan — TCP/3389 and TCP/22 listening	Services available; no established session shown	Review service exposure
F-005	PSXView false discrepancies	Anomaly leads, not proof of malware	Correlate before escalation
F-006	FTK verification — recorded match; no bad blocks	Disk acquisition integrity supported	Retain E01 + verification record

2. Investigation Context, Scope & Method5

2.1 Scenario

The investigation was performed in an isolated laboratory environment. A controlled suspicious account was created so that account-management events could be investigated and correlated with volatile and disk evidence. The objective was to demonstrate the investigative process that would be applied to a real endpoint, not to represent a real victim environment.

2.2 Scope

- Windows Security event analysis: 4624, 4720, 4732 and review of 4625 where available.
- Volatile memory acquisition and SHA-256 hashing workflow.
- Forensic disk acquisition using E01 format and verification.
- Memory analysis using Volatility 3.
- Disk and file-system analysis using Autopsy.
- Steganography hide/reveal testing using QuickStego.
- File-format and metadata comparison using ExifTool.
- Evidence tracking and chain-of-custody documentation.

2.3 Investigation method

The workflow followed an evidence-first sequence: Identify → Preserve → Acquire → Examine → Analyse → Correlate → Document → Conclude.

Analytical principle

The investigation was deliberately written to distinguish confirmed observations from hypotheses and anomaly leads. Where the retained evidence did not support a conclusion, the conclusion was qualified rather than forced.

3. Investigation Environment5

Component	Recorded configuration / observation
Target	Windows 11 virtual machine running under Oracle VirtualBox
Network posture	Host-only / isolated lab network
Analyst transfer location	VM Transfers shared folder
Memory capture	Magnet RAM Capture v1.2.0; 4,741 MB shown as available for capture
Disk acquisition	FTK Imager 4.7.1.2; segmented E01 image
Memory analysis	Volatility 3 Framework 2.28.0
Disk analysis	Autopsy 4.23.1
Steganography	QuickStego
Metadata	ExifTool 13.59

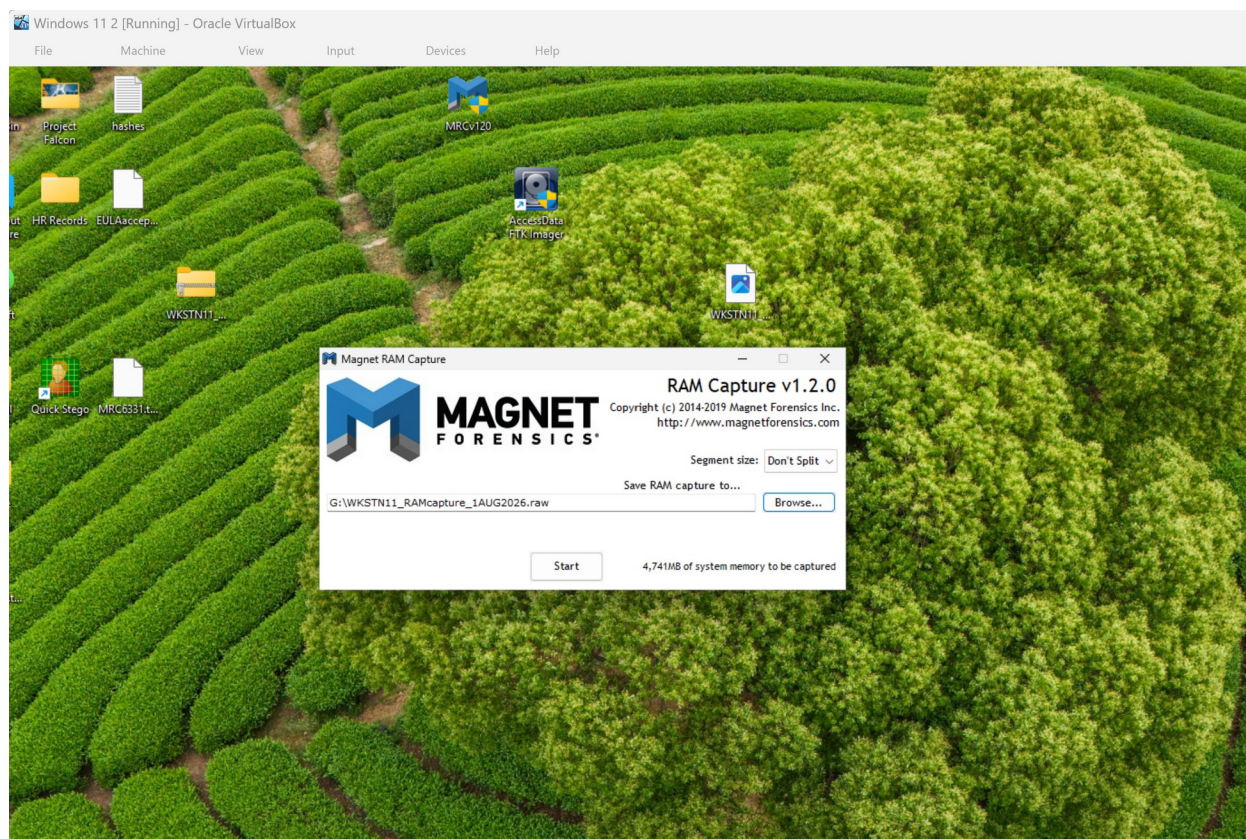


Figure 1 — Magnet RAM Capture configured to write WKSTN11_RAMcapture_1AUG2026.raw; the interface reported 4,741 MB of system memory to be captured.

4. Threat Hunting & Event Correlation⁷

4.1 Event 4720 — Account Creation

Field	Finding
Event ID	4720 — user account created
Account created	hacker_backdoor
Subject account	James_ (James_Akor)
Logged	1 August 2026, 10:39:47 AM
Assessment	Confirmed controlled account-creation event

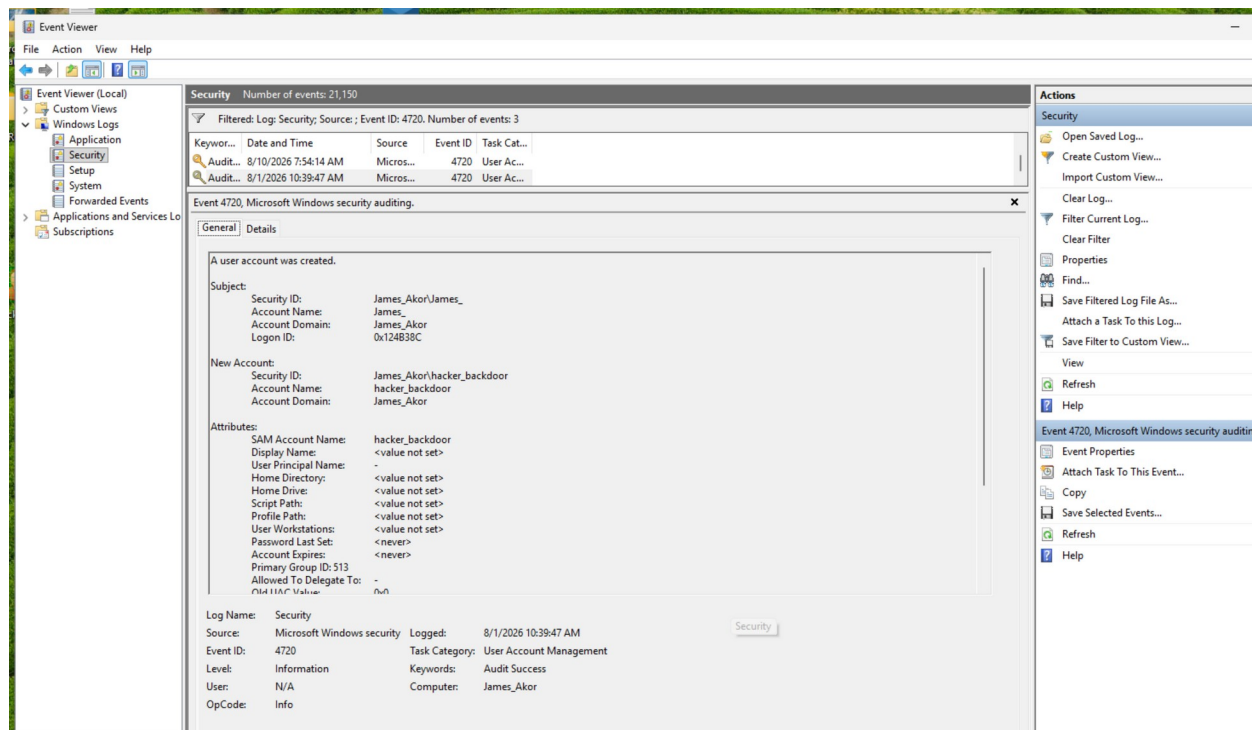


Figure 2 — Event ID 4720 showing creation of the local hacker_backdoor account.

4.2 Event 4732 — Administrator Group Membership

Field	Finding
Event ID	4732 — member added to a security-enabled local group
Member	James_Akor\hacker_backdoor
Group	BUILTIN\Administrators
Subject account	James_
Logged	1 August 2026, 10:40:41 AM
Assessment	Confirmed privilege assignment

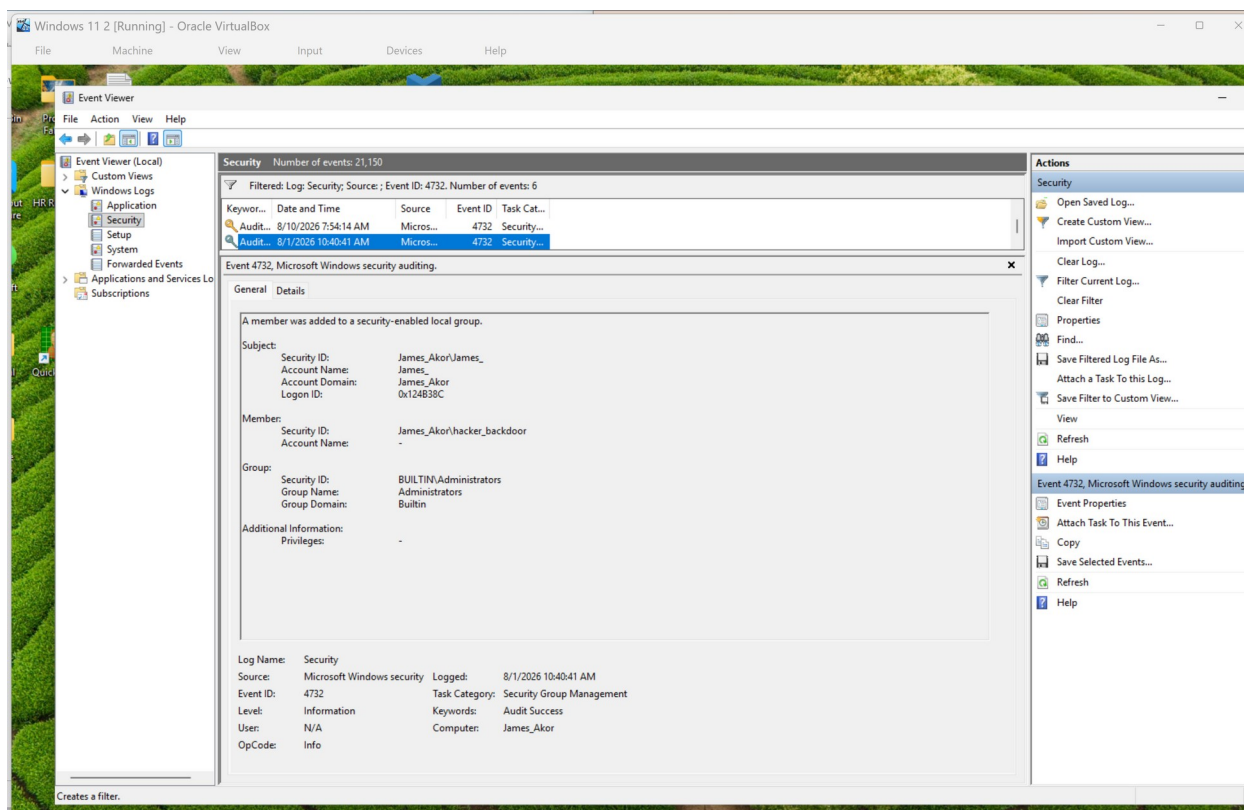


Figure 3 — Event ID 4732 showing hacker_backdoor added to BUILTIN\Administrators.

4.3 Event 4624 — Qualification of the RDP Hypothesis

The retained 4624 evidence was reviewed rather than assumed to be an RDP event. The selected event was logged at 10:40:27 AM and shows Logon Type 5, Logon Process Advapi, Process Name C:\Windows\System32\services.exe, and an IP address of “-”. This is consistent with a service logon and should not be labelled as Remote Desktop.

Field	Observed value
Event ID	4624
Selected event time	1 August 2026, 10:40:27 AM
Logon Type	5 — service logon
Logon Process	Advapi
Process	C:\Windows\System32\services.exe
IP address	-
RDP conclusion	Not confirmed by this retained event

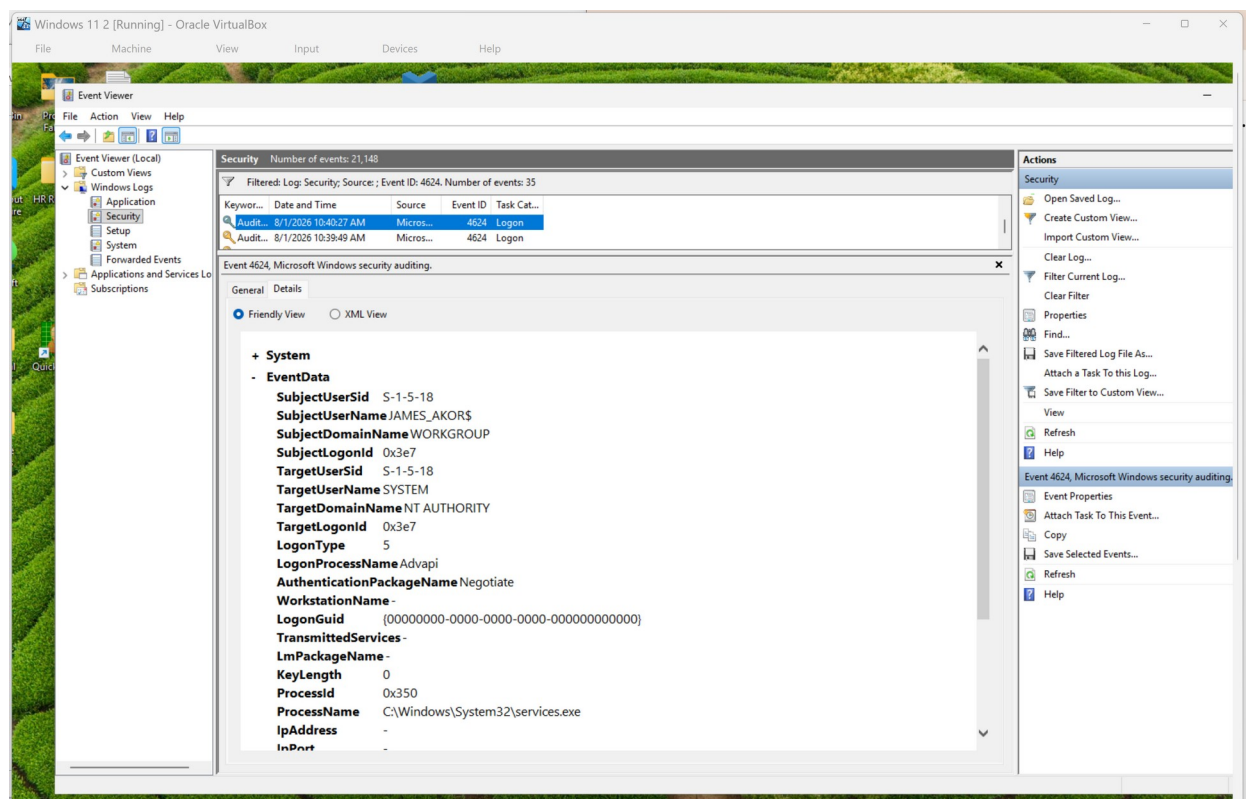


Figure 4 — Event ID 4624 details; the retained event is Logon Type 5 rather than the expected RDP Logon Type 10.

Why this matters

A strong analyst does not force evidence to fit the original hypothesis. Account creation and administrator membership are confirmed. RDP availability is corroborated later by memory analysis, but an actual Type 10 logon is not established by the retained 4624 screenshot.

4.4 Correlated Timeline

Time	Source	Observation	Interpretation
10:39:47	Security 4720	hacker_backdoor created	Confirmed account creation
10:40:27	Security 4624	Logon Type 5 / services.exe	Service logon; not RDP evidence
10:40:41	Security 4732	hacker_backdoor added to Administrators	Confirmed privilege assignment
17:51:52	Volatility netscan	TCP 3389 listening via svchost.exe	RDP service available
17:51:54–56	Volatility	sshd.exe listening on TCP 22	OpenSSH service active
17:59:46 UTC	Volatility windows.info	Memory image system time	Reference time for memory state
18:58:33 WAT	Autopsy BAM	MRCv120.exe recently executed	Corroborates memory acquisition tooling

5. Evidence Acquisition & Preservation¹⁰

5.1 RAM Acquisition

Volatile memory was captured before relying on disk-only evidence. Magnet RAM Capture v1.2.0 was configured to save WKSTN11_RAMcapture_1AUG2026.raw. A completed capture was then visible in the VM Transfers location.

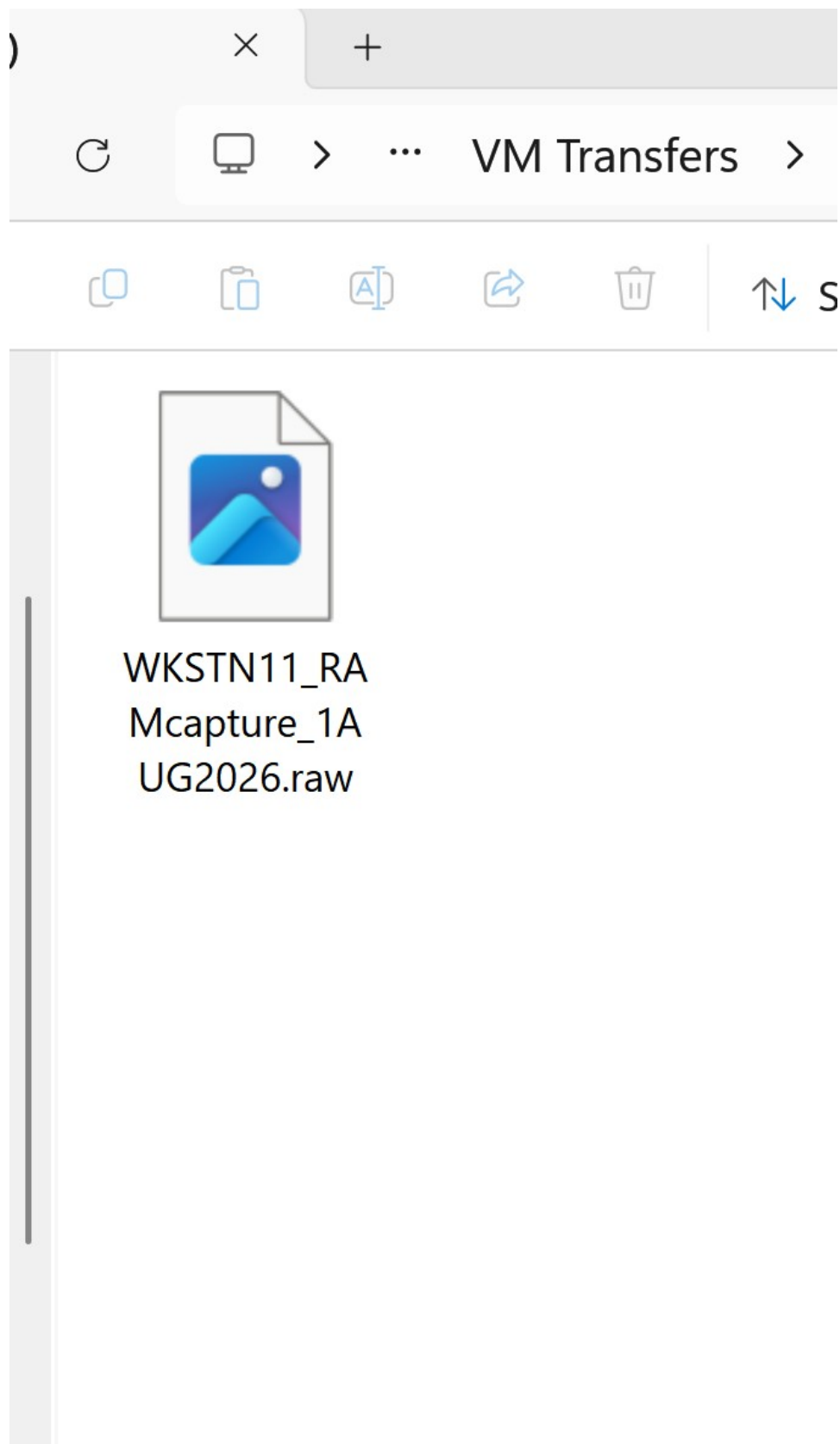


Figure 5 — Completed RAW memory capture visible in the VM Transfers location as WKSTN11_RAMcapture_1AUG2026.raw.

5.2 SHA-256 Verification

The RAW memory image was hashed with SHA-256 using certutil. The recorded digest is 51b111a9447b13d69d42b10d2d06e03324cff07cc63e4938c7f9dd8eda9305b7. The same SHA-256 value was recorded for the VM copy and the host copy after transfer, providing an integrity check across the transfer.

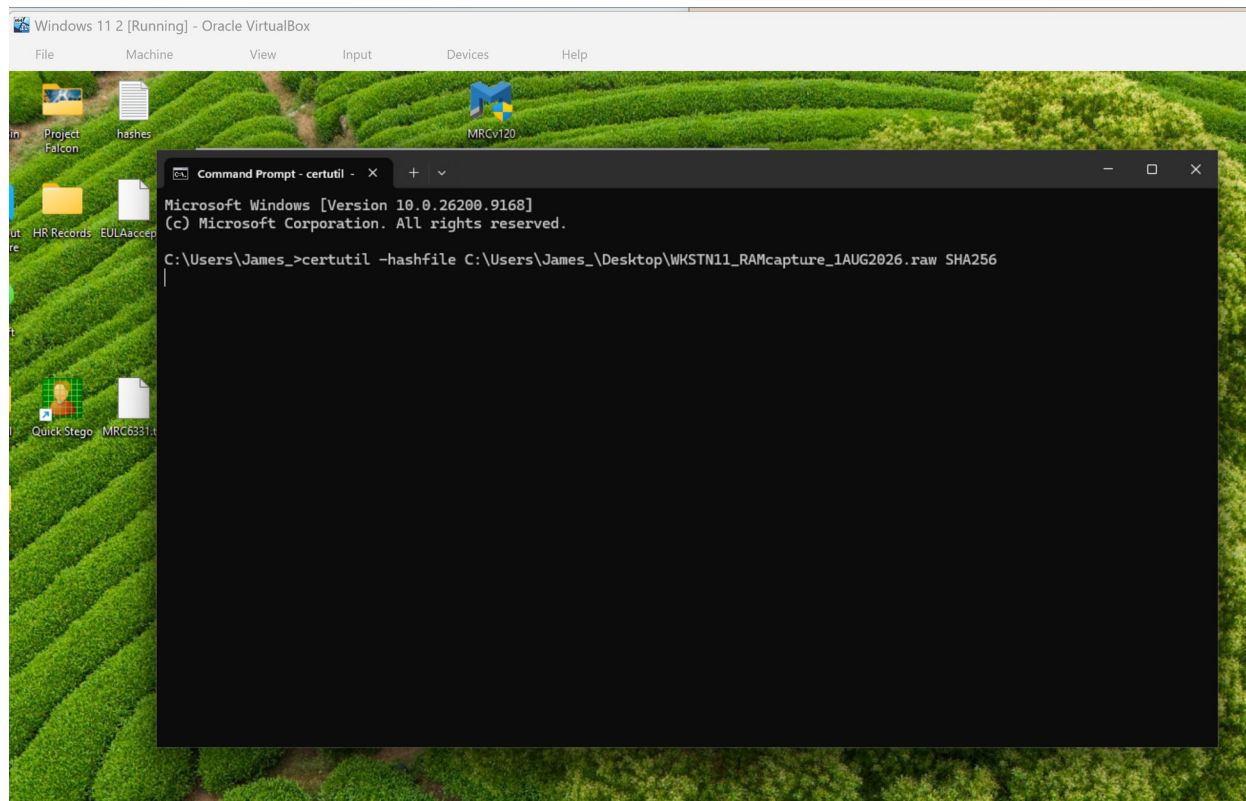


Figure 6 — SHA-256 verification command executed against the RAM capture. Recorded SHA-256: 51b111a9447b13d69d42b10d2d06e03324cff07cc63e4938c7f9dd8eda9305b7.

Evidence-quality note — The RAM SHA-256 is recorded as 51b111a9447b13d69d42b10d2d06e03324cff07cc63e4938c7f9dd8eda9305b7. The VM and host values matched after transfer. In a production investigation, the digest should be independently re-calculated and retained with the evidence.

5.3 Disk Acquisition

FTK Imager was used to create a forensic E01 image. The resulting image was segmented across E01–E31 files, with the primary evidence file named WKSTN11_DiskImage_1AUG2026.E01 and a corresponding text sidecar visible in the transfer directory.

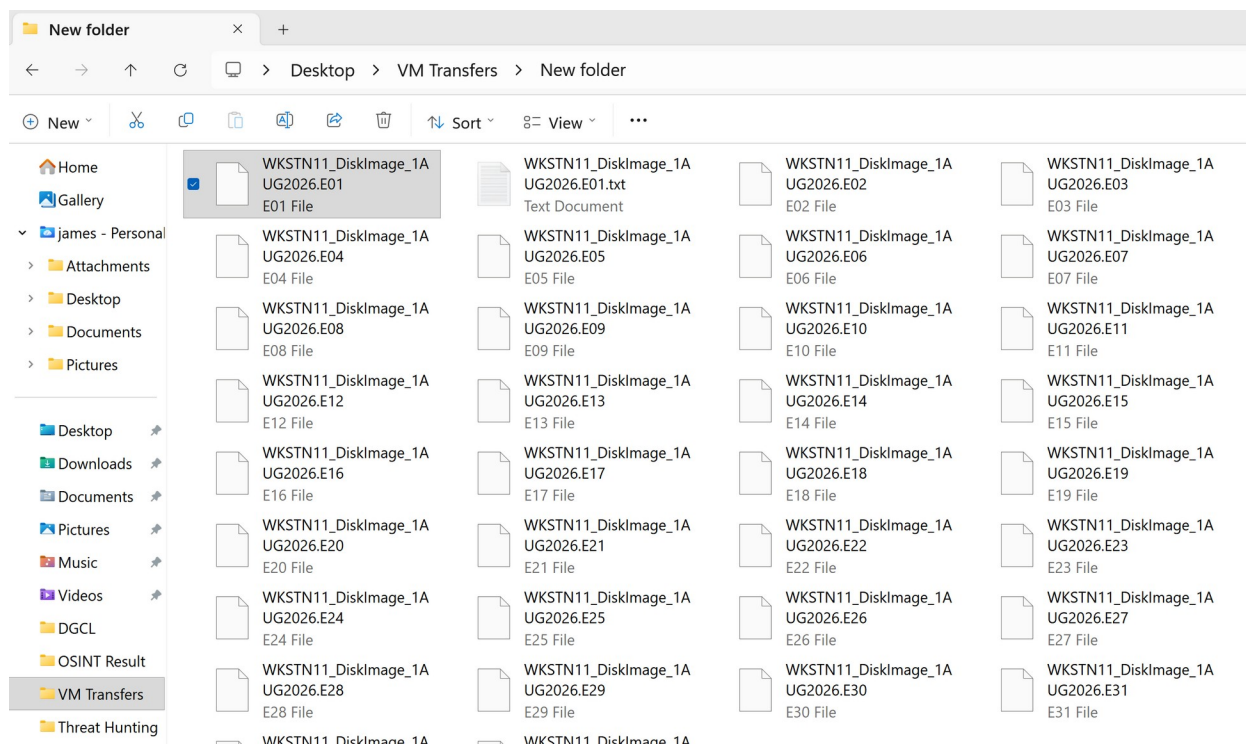


Figure 7 — Segmented E01 acquisition in VM Transfers; the primary evidence file is *WKSTN11_DiskImage_1AUG2026.E01*.

5.4 FTK Verification

FTK Imager verification reported matching computed, stored-verification and report hash records, with no bad blocks reported. The retained FTK screenshot exposes the verification fields; it should not be represented as a SHA-256 digest. The result supports the integrity of the acquired E01 image against its recorded verification values.

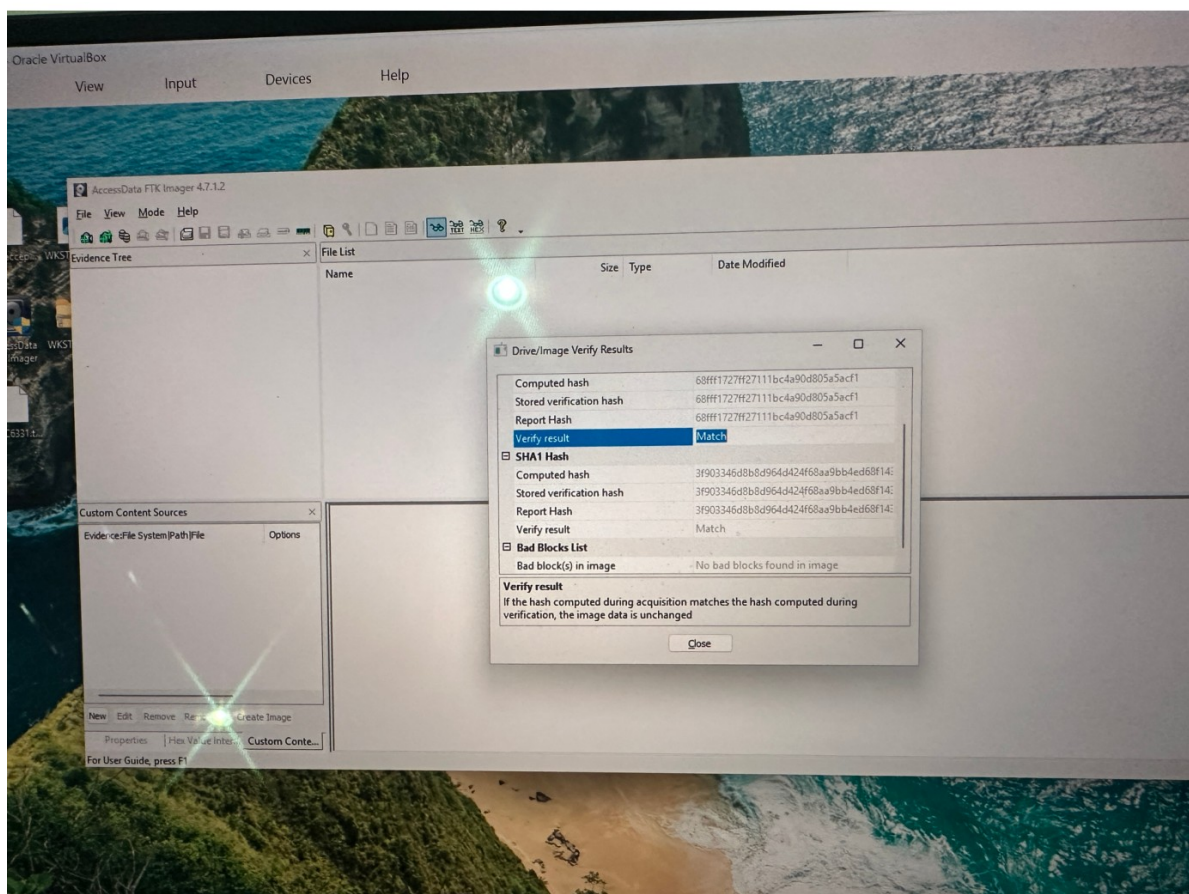


Figure 8 — FTK Imager Drive/Image Verify Results: computed, stored and report hashes matched; no bad blocks were reported.

6. Evidence Register & Chain of Custody¹⁵

The evidence register distinguishes original acquisition evidence from derived laboratory artefacts. Acquisition, transfer and verification times are recorded from the investigation notes, and the RAM SHA-256 is recorded from the retained hash record.

ID	Description	Source / Tool	Location	Integrity status	Analyst note
E-001	Volatile RAM capture	Windows 11 / Magnet RAM Capture 1.2.0	VM Transfers; originally captured as G:\WKSTN11_RAMcapture_1AUG2026.raw	SHA-256 verified; VM and host values matched	51b111a9447b13d69d42b10d2d06e03324cff07cc63e4938c7f9dd8eda9305b7
E-002	Forensic disk image	Windows 11 / FTK Imager 4.7.1.2	VM Transfers; WKSTN11_DiskImage_1AUG2026.E01 + E01–E31 segments	FTK verification: Match; no bad blocks	Retain E01 segments and FTK verification record.
E-003	image.bmp	Disk image / Autopsy	/Users/James_/Desktop/Project Falcon/image.bmp	Allocated file; metadata observed	Derived filesystem artefact; not deleted-file recovery.
E-004	fur1.jpeg / fur1.bmp	QuickStego + ExifTool	Analyst Desktop	Derived lab test artefacts	Used for hidden-message recovery and format comparison.

Evidence timeline — 1 Aug 2026: RAM acquisition 7:52 PM → RAM transfer to host 7:55 PM → disk image acquisition and FTK verification 9:35 PM → E01 loaded into Autopsy 9:45 PM.

6.1 Handling Sequence

Seq.	Evidence	From	To	Purpose
1	E-001 RAM	Target Windows VM — 1 Aug 2026, 7:52 PM	VM Transfers / analyst host — 1 Aug 2026, 7:55 PM	Preserve volatile evidence for analysis
2	E-002 E01	Target Windows VM — 1 Aug 2026, 9:35 PM	VM Transfers / analyst host — 1 Aug 2026, 9:35 PM	Create a forensic disk image
3	E-002 E01	VM Transfers — 1 Aug 2026, 9:35 PM	Autopsy case — 1 Aug 2026, 9:45 PM	Examine filesystem and deleted-file artefacts
4	E-001 RAM	VM Transfers	Volatility workstation	Memory analysis and process/network review

Chain-of-custody quality note — Recorded sequence: RAM acquired 1 Aug 2026 at 7:52 PM; transferred to the host at 7:55 PM with matching SHA-256 values; disk image acquired and verified at 9:35 PM; E01 loaded into Autopsy at 9:45 PM. The RAM SHA-256 is recorded in this report.

7. Memory Forensics with Volatility17

7.1 Tool Validation

Volatility 3 Framework 2.28.0 was installed through Python and validated with the `v -h` command. A `v.bat` wrapper was used to avoid the Windows built-in “vol” command collision described in the working guide.

```
C:\Users\DELL>v -h
usage: vol.exe [-h] [-c CONFIG]
               [--parallelism [{processes,threads,off}]] [-e EXTEND]
               [-p PLUGIN_DIRS] [-s SYMBOL_DIRS] [-v] [-l LOG]
               [-o OUTPUT_DIR] [-q] [-f FILE] [--write-config]
               [--save-config SAVE_CONFIG] [--clear-cache]
               [--cache-path CACHE_PATH] [--offline | -u URL]
               [--filters FILTERS]
               [--hide-columns [HIDE_COLUMNS ...]] [-r RENDERER]
               [--single-location SINGLE_LOCATION]
               [--stackers [STACKERS ...]]
               [--single-swap-locations [SINGLE_SWAP_LOCATIONS ...]]
               PLUGIN ...

An open-source memory forensics framework

options:
  -h, --help                Show this help message and exit, for specific
                             plugin options use 'vol.exe <pluginname>
```

Figure 9 — Volatility command wrapper validated successfully and returned the framework help output.

7.2 windows.info

Observation	Result
Framework	Volatility 3 Framework 2.28.0
Architecture	64-bit Windows
Kernel base	0xf806c2600000
Major/Minor	15.26100
Processors	4
System time	2026-08-01 17:59:46+00:00
Windows root	C:\WINDOWS
Interpretation	Memory image parsed successfully and was suitable for subsequent plugin analysis.

```

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C:\Users\DELL>v -f "C:\Users\DELL\Desktop\VM Transfers\WKSTN11_RAMcapture_1AUG2026.raw" windows.info
Volatility 3 Framework 2.28.0
Progress: 100.00      PDB scanning finished
Variable      Value

Kernel Base    0xf806c2600000
DTB            0x1ae000
Symbols file:  C:/Users/DELL/AppData/Local/Python/pythoncore-3.14-64/Lib/site-packages/volatility3/symbols/windows/ntkrnlmp.pdb/E6426A07C26E66F8EB8594621A7B273B-1.json.xz
Is64Bit       True
IsPAE         False
Layer_name     0 WindowsIntel32e
memory_layer   1 FileLayer
KdVersionBlock 0xf806c340a860
Major/Minor    15.26100
MachineType    34404
KeNumberProcessors 4
SystemTime     2026-08-01 17:59:46+00:00
NtSystemRoot   C:\WINDOWS
NtProductType  NtProductWinNt
NtMajorVersion 10
NtMinorVersion 0
PE MajorOperatingSystemVersion 10
PE MinorOperatingSystemVersion 0
PE Machine     34404
PE TimeDateStamp  Fri Mar 9 22:53:21 2085

C:\Users\DELL>

```

Figure 10 — Volatility windows.info successfully identified the Windows memory layer and system state.

7.3 Process Analysis

The retained process analysis identified MRCv120.exe, sshd.exe, explorer.exe and lsass.exe among the queried process names. MRCv120.exe is consistent with the memory-capture tool used during the exercise. sshd.exe is significant because network analysis also shows TCP/22 listening, but the retained evidence does not establish that the SSH service was malicious.

```

C:\Users\DELL>v -f "C:\Users\DELL\Desktop\VM Transfers\WKSTN11_RAMcapture_1AUG2026.raw" windows.psscan | findstr /I "MRCv120.exe sshd.exe lsass.exe explorer.exe"
936gress748100.0lsass.exe 0xb70113a5080in8shed - 0 False 2026-08-01 17:51:51.000000 UTC N/A Disabled
3940 892 sshd.exe 0xb70114abc080 2 - 0 False 2026-08-01 17:51:54.000000 UTC N/A Disabled
9756 5924 MRCv120.exe 0xb70114b5e080 4 - 1 True 2026-08-01 17:58:29.000000 UTC N/A Disabled
5924 5896 explorer.exe 0xb7011576b080 49 - 1 False 2026-08-01 17:52:22.000000 UTC N/A Disabled

C:\Users\DELL>

```

Figure 11 — Volatility psscan output showing the capture utility, OpenSSH and normal Windows processes in memory.

7.4 Network Analysis

Volatility netstat showed the system listening on TCP 3389 through svchost.exe (PID 1348) over IPv4 and IPv6, and on TCP 22 through sshd.exe (PID 3940). The retained output shows listening sockets rather than an established remote connection.

```

C:\Users\DELL>v -f "C:\Users\DELL\Desktop\VM Transfers\WKSTN11_RAMcapture_1AUG2026.raw" windows.netstat | findstr /I "sshd.exe 3389"
0xb7010e4f8488.0TCPv4 0.0.0.0 22B scan0.0.0.0n0shed LISTENING 3940 sshd.exe 2026-08-01 17:51:56.000000 UTC
0xb7010e4f8488 TCPv6 :: 22 :: 0 LISTENING 3940 sshd.exe 2026-08-01 17:51:56.000000 UTC
0xb7010e4f8cc8 TCPv4 0.0.0.0 22 0.0.0.0 0 LISTENING 3940 sshd.exe 2026-08-01 17:51:56.000000 UTC
0xb7011310d248 TCPv4 0.0.0.0 3389 0.0.0.0 0 LISTENING 1348 svchost.exe 2026-08-01 17:51:52.000000 UTC
0xb7011310d248 TCPv6 :: 3389 :: 0 LISTENING 1348 svchost.exe 2026-08-01 17:51:52.000000 UTC
0xb701139e8508 TCPv4 0.0.0.0 3389 0.0.0.0 0 LISTENING 1348 svchost.exe 2026-08-01 17:51:52.000000 UTC
0xb701140bd2b8 UDPv4 0.0.0.0 3389 * 0 1348 svchost.exe 2026-08-01 17:51:52.000000 UTC
0xb701140bd2b8 UDPv6 :: 3389 * 0 1348 svchost.exe 2026-08-01 17:51:52.000000 UTC
0xb701140bde18 UDPv4 0.0.0.0 3389 * 0 1348 svchost.exe 2026-08-01 17:51:52.000000 UTC

C:\Users\DELL>

```

Figure 12 — Volatility network evidence: TCP/3389 and TCP/22 listening sockets were present; no established session is shown in this filtered output.

7.5 PSXView Review

The PSXView command was filtered for “False” results, producing processes with discrepancies across process enumeration methods. This is an anomaly indicator, not a standalone malware verdict. The output warrants correlation with process paths, signatures, network activity and other artefacts before escalation.

```
C:\Users\DELL>v -f "C:\Users\DELL\Desktop\VM Transfers\Wkstn11_RAMcapture_1AUG2026.raw" windows.malware.psxview | findstr /I "False"
0xb7011487b080.0audiodg.exe 3400scanFalsefinTrued False False 2026-08-01 18:03:37+00:00
0xb7010b697040 System 4 True False True False False
0xb7010efef040 smss.exe 504 True True True False
0xb7010b69f040 MemCompression 2768 True True True False
0xb701155cd080 svchost.exe 5600 False True False False 2026-08-01 18:04:21+00:00
0xb7011648f080 WindowsTermina 3088 False True False False 2026-08-01 17:58:31+00:00
0xb7010b73e080 Registry 128 True True True False
0xb7011399f080 svchost.exe 5520 True True True False 2026-08-01 18:07:14+00:00
0xb70114a9f080 WmiPrvSE.exe 7544 False True False False 2026-08-01 18:03:54+00:00
0xb70115c80080 svchost.exe 5784 True True False False 2026-08-01 17:56:17+00:00
0xb70113c290c0 WerFault.exe 916 True True False False 2026-08-01 17:52:27+00:00
0xb7011490c080 SearchFilterHo 6716 False True False False 2026-08-01 17:54:48+00:00
0xb70113e52080 svchost.exe 1336 False True False False 2026-08-01 18:01:46+00:00
0xb70114281080 svchost.exe 8416 False True True False 2026-08-01 18:07:54+00:00
0xb70113306140 csrss.exe 676 True True True False
0xb70113a63140 csrss.exe 756 True True True False
0xb701166cf080 svchost.exe 10060 False True False False 2026-08-01 17:54:02+00:00
0xb7010b7b0080 svchost.exe 5700 False True False False 2026-08-01 17:59:14+00:00
0xb70115648080 userinit.exe 5896 True True False False 2026-08-01 17:52:54+00:00
0xb70114e2d080 MolsoCoreWorke 3652 False True True False 2026-08-01 18:07:56+00:00
0xb70114c73080 smartscreen.ex 8644 False True True False 2026-08-01 18:07:44+00:00
0xb701163f3080 sppsvc.exe 10096 False True False False 2026-08-01 17:54:37+00:00
C:\Users\DELL>
```

Figure 13 — Volatility PSXView output filtered for false results. The discrepancies were treated as leads requiring correlation, not automatic malware findings.

8. Disk Forensics with Autopsy20

8.1 Case and Image

Autopsy 4.23.1 was used to create the Module 10 case and ingest WKSTN11_DiskImage_1AUG2026.E01. The data source is visible in the case as WKSTN11_DiskImage_1AUG2026.E01_1 Host.

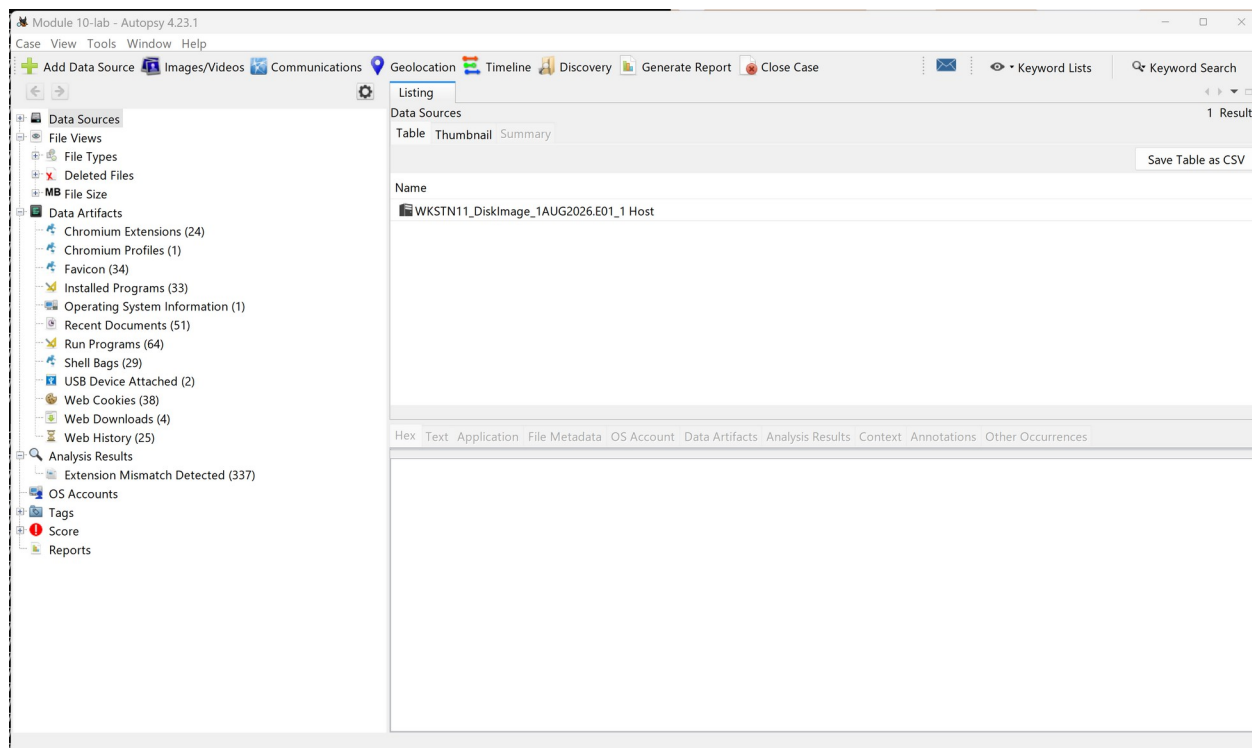


Figure 14 — Autopsy case with the acquired E01 data source loaded.

8.2 Main Windows Partition

The main Windows volume was identified as vol6, a Basic data partition spanning sectors 444416–166262783. The path displayed in Autopsy shows the Windows user profile under /Users/James_. This is the partition used for the principal filesystem review.

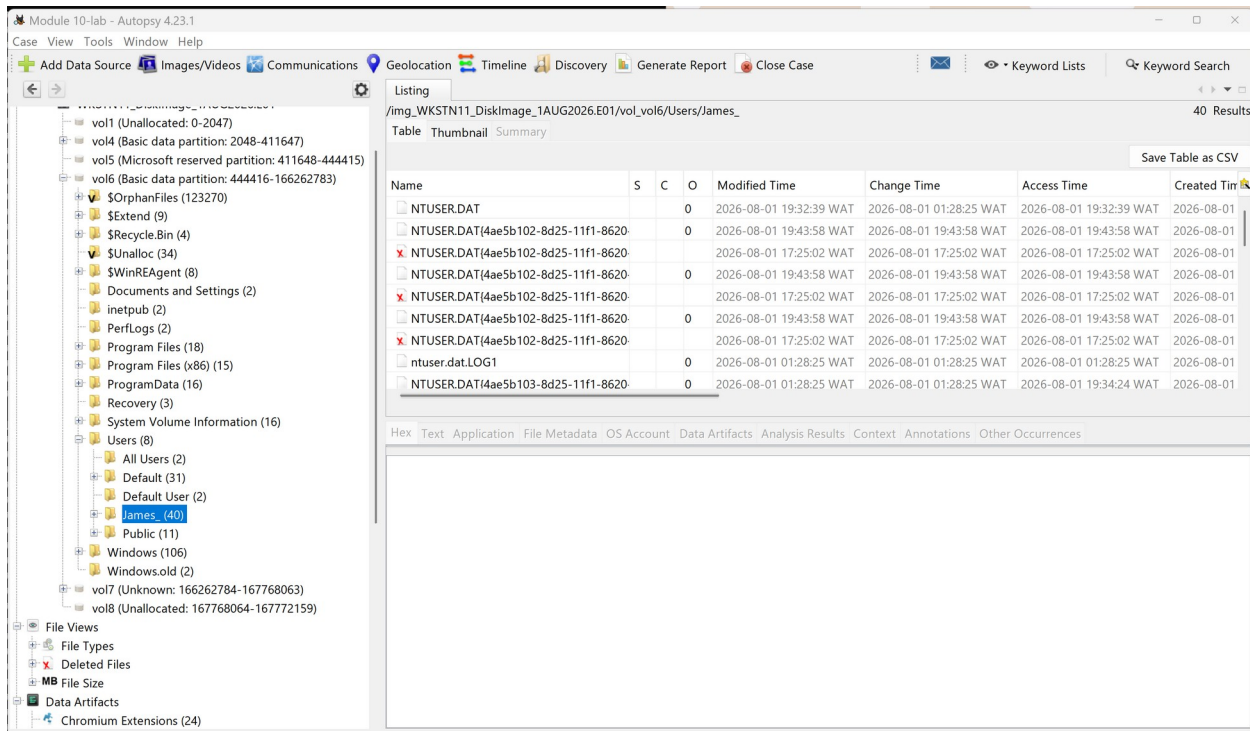


Figure 15 — Autopsy main Windows partition (vol6) with the James_ user profile visible.

8.3 Deleted-File Review

Autopsy reported 1,100,291 entries under Deleted Files / File System and the same total under Deleted Files / All. Sorting by size showed that many of the largest results were Windows/application artefacts such as content.bin, vmfirmwarecl.dll and WindowsOobeAppHost.AOT.dll.

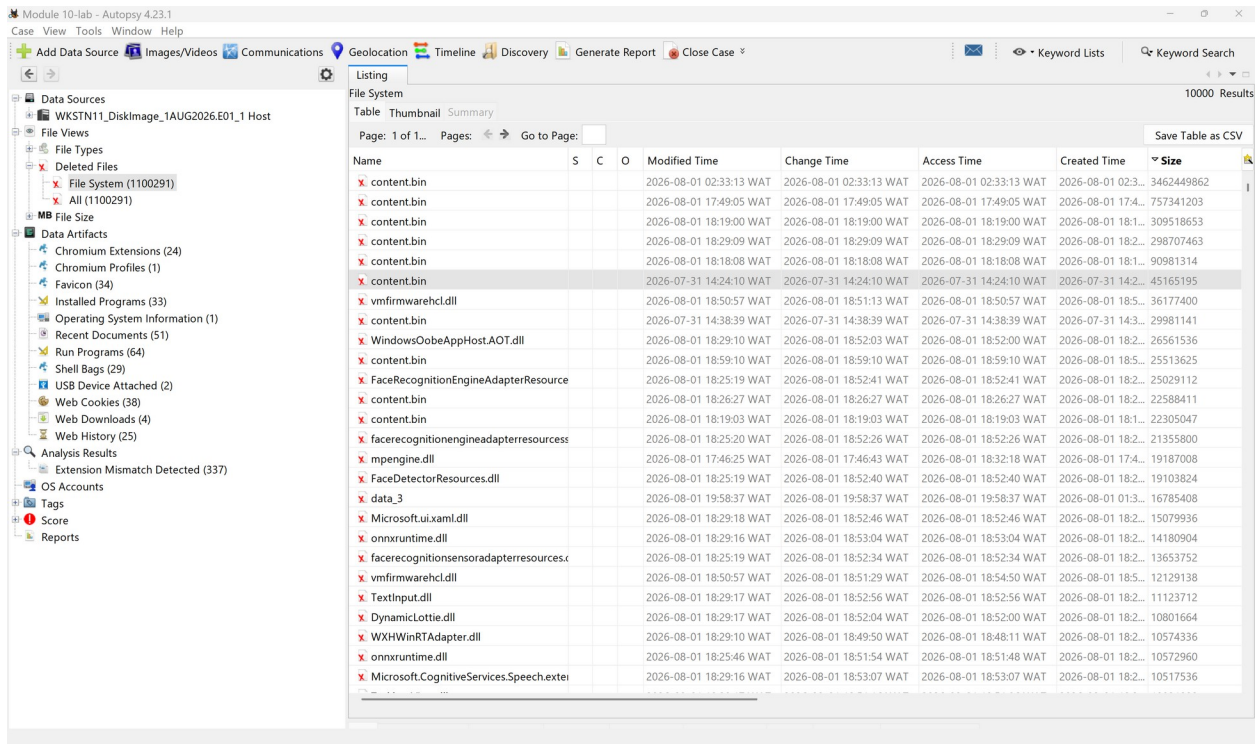


Figure 16 — Autopsy deleted-file results sorted by size; the retained screenshot is dominated by system/application artefacts.

Finding qualification

The retained evidence does not establish a user document as a notable deleted-file finding. The large count should therefore not be presented as proof of user-file deletion.

8.4 BAM Execution Artefact

A separate Autopsy artefact review identified MRCv120.exe at \Users\James_\Desktop\MRCv120.exe. The Windows Registry Analysis / Background Activity Moderator (BAM) entry records the program as recently executed at 2026-08-01 18:58:33 WAT. This corroborates that the memory-capture utility was executed on the endpoint.

The screenshot shows the Autopsy 4.23.1 interface. The left sidebar displays the file tree, with 'Run Programs (64)' selected under 'Data Artifacts'. The main pane shows a table of executed programs. The table has columns: Source Name, S, C, O, Program Name, Username, Date/Time, and Comment. The entry for 'MRCv120.exe' is highlighted, showing it was executed at 2026-08-01 18:58:33 WAT by user S-1-5-90-0-1. Below the table, a detailed view of the selected artefact is shown, including fields like Type, Program Name, Username, Date/Time, Comment, Source File Path, and Artifact ID.

Source Name	S	C	O	Program Name	Username	Date/Time	Comment
SYSTEM				\\Windows\\System32\\WindowsPowerShell...		2026-08-01 19:26:32 WAT	Recently Executed according to
SYSTEM				Microsoft.SecHealthUI_8wekyb3d8bbwe		2026-08-01 19:26:41 WAT	Recently Executed according to
SYSTEM				\\Windows\\System32\\Taskmgr.exe		2026-08-01 02:42:45 WAT	Recently Executed according to
SYSTEM				Microsoft.Windows.ShellExperienceHost_c...		2026-08-01 19:32:37 WAT	Recently Executed according to
SYSTEM				Microsoft.XboxGamingOverlay_8wekyb3d8...		2026-08-01 17:52:32 WAT	Recently Executed according to
SYSTEM				\\Windows\\System32\\OpenWith.exe		2026-08-01 18:58:14 WAT	Recently Executed according to
SYSTEM				Microsoft.WindowsNotepad_8wekyb3d8bb...		2026-08-01 19:26:47 WAT	Recently Executed according to
SYSTEM				\\Windows\\System32\\DataExchangeHost.exe		2026-08-01 19:10:39 WAT	Recently Executed according to
SYSTEM				\\Program Files (x86)\\Microsoft\\Edge\\Appli...		2026-08-01 19:46:28 WAT	Recently Executed according to
SYSTEM				Microsoft.LockApp_cw5n1h2xyewy		2026-08-01 20:06:55 WAT	Recently Executed according to
SYSTEM				\\Program Files\\Oracle\\VirtualBox Guest Ad...		2026-08-01 19:27:53 WAT	Recently Executed according to
SYSTEM				\\Windows\\System32\\msiexec.exe		2026-08-01 12:11:49 WAT	Recently Executed according to
SYSTEM				\\Program Files (x86)\\Quick Stego\\quickste...		2026-08-01 12:18:10 WAT	Recently Executed according to
SYSTEM				\\Users\\James_\\Desktop\\MRCv120.exe		2026-08-01 18:58:33 WAT	Recently Executed according to
SYSTEM				\\Windows\\System32\\dwm.exe	S-1-5-90-0-1	2026-08-02 03:43:40 WAT	Recently Executed according to

Type	Value	Source(s)
Program Name	\\Users\\James_\\Desktop\\MRCv120.exe	Windows Registry Analyz
Username		Windows Registry Analyz
Date/Time	2026-08-01 18:58:33 WAT	Windows Registry Analyz
Comment	Recently Executed according to Background Activity Moderator (BAM)	Windows Registry Analyz
Source File Path	/img_WKSTN11_DiskImage_1AUG2026.E01/vol_vol6/Windows/System32/config/SYSTEM	
Artifact ID	-9223372036854775244	

Figure 17 — Autopsy BAM artefact showing recent execution of MRCv120.exe from the James_ Desktop.

8.5 Located Image Artefact

Autopsy also located image.bmp under /Users/James_/Desktop/Project Falcon/image.bmp. The file is shown as allocated and approximately 3.81 MB. Because the file is allocated, it should not be described as a deleted-file recovery.

Module 10-lab - Autopsy 4.23.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Listing image/bmp 10 Results

Data Sources

- WKSTN11_DiskImage_1AUG2026.E01_1 Host
 - WKSTN11_DiskImage_1AUG2026.E01
 - vol1 (Unallocated: 0-2047)
 - vol4 (Basic data partition: 2048-411647)
 - vol5 (Microsoft reserved partition: 411648-444415)
 - vol6 (Basic data partition: 444416-166262783)
 - vol7 (Unknown: 166262784-167768063)
 - vol8 (Unallocated: 167768064-167772159)

File Views

- By Extension
 - By MIME Type
 - application
 - audio
 - image
 - vnd.microsoft.icon
 - bmp
 - gif
 - x-portable-floatmap
 - png
 - jpeg
 - svg+xml
 - vnd.zbrush.pcx
 - webp
 - avif

Deleted Files

 - File System (1100291)
 - All (1100291)

MB File Size

Data Artifacts

 - Chromium Extensions (24)
 - Chromium Profiles (1)
 - Favicon (34)
 - Installed Programs (33)
 - Operating System Information (1)

Table Thumbnail Summary

Page: 1 of 1 Pages: Go to Page: Save Table as CSV

| Name | S | C | O | Modified Time | Change Time | Access Time | Created Time | Size | Flags(Dir) | F |
|--------------------|---|---|---|-------------------------|-------------------------|-------------------------|-------------------------|---------|-------------|---|
| guest.bmp | | | 0 | 2024-04-01 08:24:04 WAT | 2026-07-31 22:55:14 WAT | 2024-04-01 08:24:04 WAT | 2024-04-01 08:26:11 WAT | 401464 | Allocated | A |
| user.bmp | | | 0 | 2024-04-01 08:24:04 WAT | 2026-07-31 22:55:14 WAT | 2024-04-01 08:24:04 WAT | 2024-04-01 08:26:11 WAT | 401464 | Allocated | A |
| image.bmp | | | 0 | 2026-08-01 12:18:01 WAT | 2026-08-01 12:18:01 WAT | 2026-08-01 19:34:45 WAT | 2026-08-01 12:18:00 WAT | 3808854 | Allocated | A |
| wmpnss_color32.bmp | | | 0 | 2024-03-28 13:50:00 WAT | 2026-08-01 19:27:29 WAT | 2024-04-01 09:02:48 WAT | 2024-04-01 09:02:48 WAT | 4152 | Allocated | A |
| wmpnss_color48.bmp | | | 0 | 2024-03-28 13:50:00 WAT | 2026-08-01 19:27:29 WAT | 2024-04-01 09:02:48 WAT | 2024-04-01 09:02:48 WAT | 9272 | Allocated | A |
| wmpnss_color32.bmp | | | 0 | 2024-03-28 13:50:00 WAT | 2026-08-01 19:27:29 WAT | 2024-04-01 09:02:48 WAT | 2024-04-01 09:02:48 WAT | 4152 | Allocated | A |
| wmpnss_color48.bmp | | | 0 | 2024-03-28 13:50:00 WAT | 2026-08-01 19:27:29 WAT | 2024-04-01 09:02:48 WAT | 2024-04-01 09:02:48 WAT | 9272 | Allocated | A |
| meiryo_boot.ttf | | | | 2026-02-03 19:39:02 WAT | 2026-08-01 18:52:07 WAT | 2026-08-01 18:25:56 WAT | 2026-08-01 18:23:29 WAT | 3414 | Unallocated | U |

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Result: 1 of 2 Result

Associated Object

| Type | Value | Source(s) |
|---------------------|---|-----------------|
| Associated Artifact | -9223372036854775447 | Recent Activity |
| Source File Path | /img_WKSTN11_DiskImage_1AUG2026.E01/vol6/Users/James/Desktop/Project Falcon/image.bmp | |
| Artifact ID | -9223372036854775446 | |

Figure 18 — Autopsy locating image.bmp in the Project Falcon folder; the file is allocated, not deleted.

9. Steganography & Metadata Analysis24

9.1 QuickStego

QuickStego was used to embed a controlled message in an image and then reveal it. The recovered message was “A fake set of credentials”. The message is intentionally synthetic and exists only for the controlled laboratory demonstration.

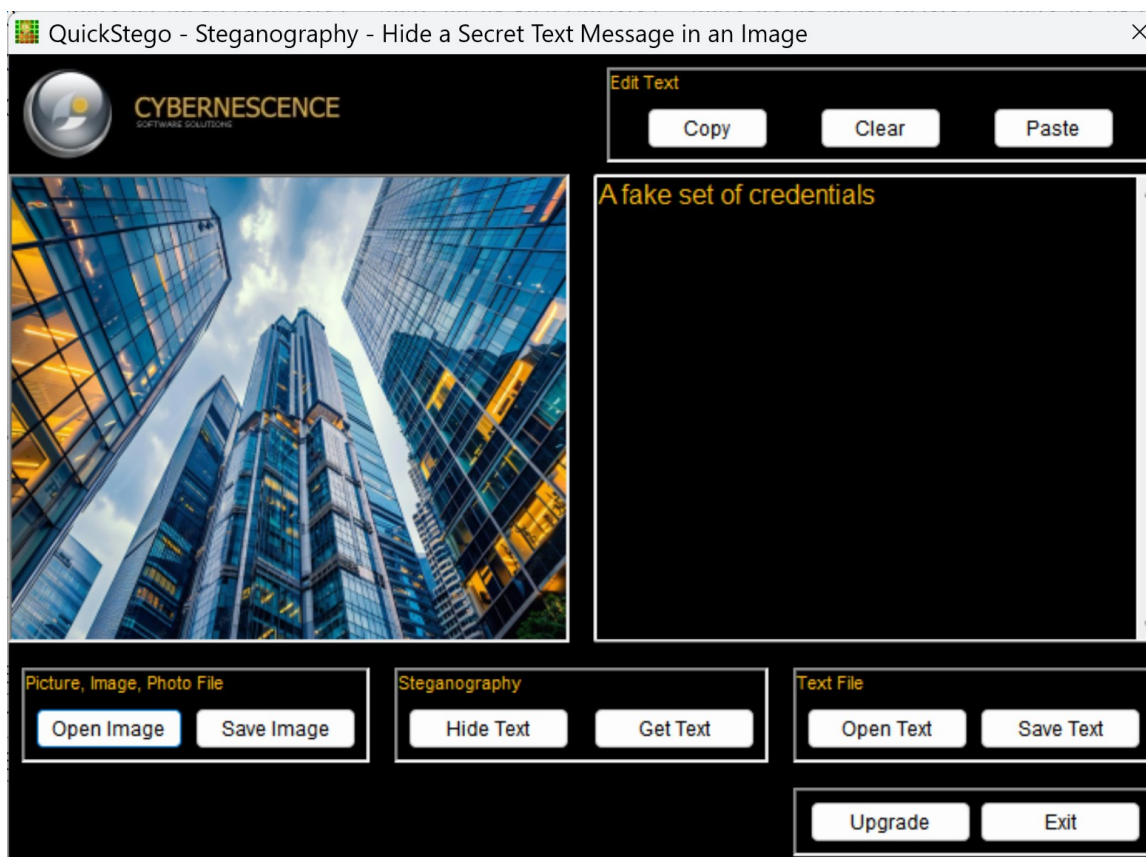


Figure 19 — QuickStego showing the recovered hidden message: “A fake set of credentials”.

9.2 ExifTool Comparison

| Attribute | fur1.jpeg | fur1.bmp |
|------------------------|---------------------------------|---------------------|
| File size | 147 KB | 3.7 MB |
| File type | JPEG | BMP |
| MIME type | image/jpeg | image/bmp |
| Dimensions | 1280 × 960 | 1280 × 960 |
| Compression / encoding | Progressive DCT, Huffman coding | None |
| Bit depth | 8 bits/sample | 24-bit |
| Image size | 1280 × 960 / 1.2 MP | 1280 × 960 / 1.2 MP |


```

C:\Users\DELL>exiftool C:\Users\DELL\Desktop\fur1.jpeg
ExifTool Version Number      : 13.59
File Name                    : fur1.jpeg
Directory                   : C:/Users/DELL/Desktop
File Size                    : 147 kB
Zone Identifier              : Exists
File Modification Date/Time   : 2026:07:08 11:15:33+01:00
File Access Date/Time        : 2026:08:16 20:49:58+01:00
File Creation Date/Time      : 2026:07:08 11:15:33+01:00
File Permissions             : -rw-rw-rw-
File Type                    : JPEG
File Type Extension          : jpg
MIME Type                    : image/jpeg
JFIF Version                 : 1.01
Resolution Unit              : None
X Resolution                 : 1
Y Resolution                 : 1
Image Width                  : 1280
Image Height                 : 960
Encoding Process              : Progressive DCT, Huffman coding
Bits Per Sample              : 8
Color Components              : 3
Y Cb Cr Sub Sampling         : YCbCr4:2:0 (2 2)
Image Size                   : 1280x960
Megapixels                   : 1.2

C:\Users\DELL>exiftool C:\Users\DELL\Desktop\fur1.bmp
ExifTool Version Number      : 13.59
File Name                    : fur1.bmp
Directory                   : C:/Users/DELL/Desktop
File Size                    : 3.7 MB
File Modification Date/Time   : 2026:08:02 07:20:28+01:00
File Access Date/Time        : 2026:08:16 20:48:22+01:00
File Creation Date/Time      : 2026:08:02 07:20:28+01:00
File Permissions             : -rw-rw-rw-
File Type                    : BMP
File Type Extension          : bmp
MIME Type                    : image/bmp
BMP Version                  : Windows V3
Image Width                  : 1280
Image Height                 : 960
Planes                      : 1
Bit Depth                    : 24
Compression                  : None
Image Length                 : 3686400
Pixels Per Meter X           : 0
Pixels Per Meter Y           : 0
Num Colors                   : Use BitDepth
Num Important Colors          : All
Image Size                   : 1280x960
Megapixels                   : 1.2

```

Figure 20 — ExifTool comparison of the original JPEG and BMP steganography test artefact.

The comparison demonstrates that hidden-content testing can produce meaningful file-format and metadata differences without changing the displayed image dimensions. The increase from 147 KB to approximately 3.7 MB and the change from JPEG compression to an uncompressed 24-bit BMP are consistent with the controlled conversion used for the exercise. These differences alone should not be interpreted as proof of malicious activity.

10. Evidence Correlation & Analyst Assessment²⁶

| Evidence source | Confirmed observation | Correlation | Analytical significance |
|--------------------|---|--|--|
| Windows 4720 | hacker_backdoor created at 10:39:47 | Correlates with later privilege assignment | Strong account-management indicator |
| Windows 4732 | hacker_backdoor added to BUILTIN\Administrators at 10:40:41 | Follows account creation by ~54 seconds | Privilege escalation / persistence lead |
| Windows 4624 | Selected event is Logon Type 5 | Does not prove RDP | Prevents overstatement of RDP hypothesis |
| Volatility netscan | 3389 listening via svchost; 22 listening via sshd | Supports service availability | Useful corroboration, not proof of remote use |
| Volatility psscan | MRCv120, sshd, explorer, lsass observed | Matches acquisition environment | Provides memory-state context |
| Volatility PSXView | Multiple false discrepancies | Requires further correlation | Anomaly lead; not a standalone malware finding |
| Autopsy BAM | MRCv120 recently executed | Corroborates RAM capture activity | Connects disk artefact to acquisition workflow |
| QuickStego | Hidden message recovered | Directly demonstrates concealment/recovery | Confirms steganography workflow |
| ExifTool | JPEG/BMP metadata and size differ | Supports controlled file transformation | Demonstrates metadata-based comparison |

10.1 Severity / Disposition

Recommended disposition: ESCALATE FOR REVIEW

In a real enterprise, an unexpected local account followed by immediate membership in the local Administrators group would justify investigation and containment review. The evidence should be correlated with identity records, endpoint management data, authentication logs and network telemetry before declaring compromise.

11. Final Investigation Conclusion²⁷

The investigation successfully demonstrated an end-to-end DFIR workflow against a controlled Windows endpoint. The most significant confirmed security finding was the creation of `hacker_backdoor` followed by its addition to `BUILTIN\Administrators`. This sequence was recovered directly from Windows Security events and is sufficient to establish suspicious account-management activity within the simulated case.

The evidence-preservation stage was also successfully demonstrated. Volatile memory was captured to a named RAW file, and a segmented E01 disk image was created and verified. FTK Imager reported matching verification records and no bad blocks. Volatility parsed the memory image, identified the Windows environment, enumerated processes and network listeners, and exposed PSXView discrepancies for further review.

Disk examination with Autopsy established the main Windows partition, quantified the large deleted-file result set and provided a BAM artefact showing recent execution of the memory-capture utility. The steganography stage successfully recovered a hidden synthetic credential message, while ExifTool demonstrated metadata and file-structure differences between the original JPEG and BMP test artefacts.

Evidence quality note — The RAM integrity record is complete: the VM and host SHA-256 values both equal `51b111a9447b13d69d42b10d2d06e03324cff07cc63e4938c7f9dd8eda9305b7`. FTK verification separately supports the E01 acquisition, with matching verification records and no bad blocks.

Professional takeaway

The value of this investigation is not that a tool was opened successfully. It is that independent evidence sources were connected, evidence integrity was considered, findings were qualified, and conclusions were limited to what the artefacts actually support.

12. Operational Recommendations²⁷

1. Validate the business owner and legitimacy of `hacker_backdoor` immediately in a real environment; if unauthorized, disable or contain the account while preserving the relevant logs and evidence.
2. Correlate authentication telemetry for the same time window, including 4624 Logon Type 10, 4778/4779 Remote Desktop session events, and endpoint/network telemetry, before concluding that an RDP session occurred.
3. Review why OpenSSH was listening on TCP/22 and confirm whether the service is required. If it is not required, remove or restrict the exposure and document the control decision.
4. The RAM SHA-256 is recorded as `51b111a9447b13d69d42b10d2d06e03324cff07cc63e4938c7f9dd8eda9305b7`; the VM and host values matched after transfer. In a production investigation, the same verification should be independently repeated and preserved with the evidence.
5. Treat PSXView discrepancies as investigative leads. Validate suspicious entries against process paths, signatures, parent/child relationships, persistence locations and network activity rather than labelling them as malware from PSXView alone.
6. Preserve the FTK verification report and E01 segment set together, maintain consistent evidence identifiers, and keep the chain-of-custody record with the case materials.

13. Analyst Reflection & Skills Demonstrated²⁸

This case strengthened my ability to move from a security alert or suspicious artefact to a documented investigative conclusion. The work required me to preserve volatile evidence before relying on disk evidence, verify forensic imaging, use memory and filesystem tools, and distinguish confirmed facts from investigative leads.

- Windows Security event analysis and event correlation
- Account-management investigation and privilege-assignment analysis
- Volatile memory acquisition and integrity workflow
- Forensic disk imaging and FTK verification
- Volatility 3 memory analysis
- Process and network artefact interpretation
- PSXView anomaly triage
- Autopsy filesystem and deleted-file analysis
- Windows BAM execution artefact analysis
- Steganography detection and hidden-message recovery
- ExifTool metadata and file-format comparison
- Evidence tracking, chain of custody and defensible reporting
- Analytical restraint: documenting uncertainty instead of overstating evidence

Appendix A — Evidence Index²⁹

| Evidence ID | Evidence / screenshot | Purpose | Key result |
|-------------|---------------------------------------|---------------------------------|--|
| EV-01 | Event 4720 — Account Creation | Confirm account creation | hacker_backdoor created at 10:39:47 AM |
| EV-02 | Event 4732 — Administrator Membership | Confirm privilege assignment | Added to BUILTIN\Administrators at 10:40:41 AM |
| EV-03 | Event 4624 — Logon Details | Test RDP hypothesis | Logon Type 5; RDP not confirmed |
| EV-04 | RAM Capture | Preserve volatile state | RAW capture created |
| EV-05 | RAM SHA-256 command | Integrity workflow | SHA-256 recorded;
VM/host match
(51b111a9447b13d69d42b10d2d06e03324cff07cc63e4938c7f9dd8eda9305b7) |
| EV-06 | FTK E01 acquisition | Preserve disk evidence | Segmented E01 image created |
| EV-07 | FTK verification | Validate image integrity | Match; no bad blocks |
| EV-08 | Volatility windows.info | Validate memory image | Windows memory parsed successfully |
| EV-09 | Volatility psscan | Review processes | MRCv120, sshd, explorer and other processes observed |
| EV-10 | Volatility netscan | Review network state | TCP/3389 and TCP/22 listening |
| EV-11 | Volatility PSXView | Triage process discrepancies | Multiple false discrepancies require correlation |
| EV-12 | Autopsy case / E01 | Confirm forensic case setup | E01 loaded successfully |
| EV-13 | Autopsy main partition | Identify Windows filesystem | vol6 / Users / James_ |
| EV-14 | Autopsy deleted files | Review deleted artefacts | 1,100,291 entries; mostly system/application artefacts in retained view |
| EV-15 | Autopsy BAM | Confirm execution artefact | MRCv120 recently executed at 18:58:33 WAT |
| EV-16 | Autopsy image.bmp | Locate filesystem artefact | Allocated Project Falcon image.bmp |
| EV-17 | QuickStego | Recover hidden content | “A fake set of credentials” |
| EV-18 | ExifTool comparison | Compare metadata/file structure | JPEG vs BMP differences documented |

Appendix B — Selected Evidence Screenshots²⁹

The following evidence figures are retained from the investigation evidence set. Installation and download screenshots are intentionally omitted because they demonstrate setup rather than investigative reasoning.

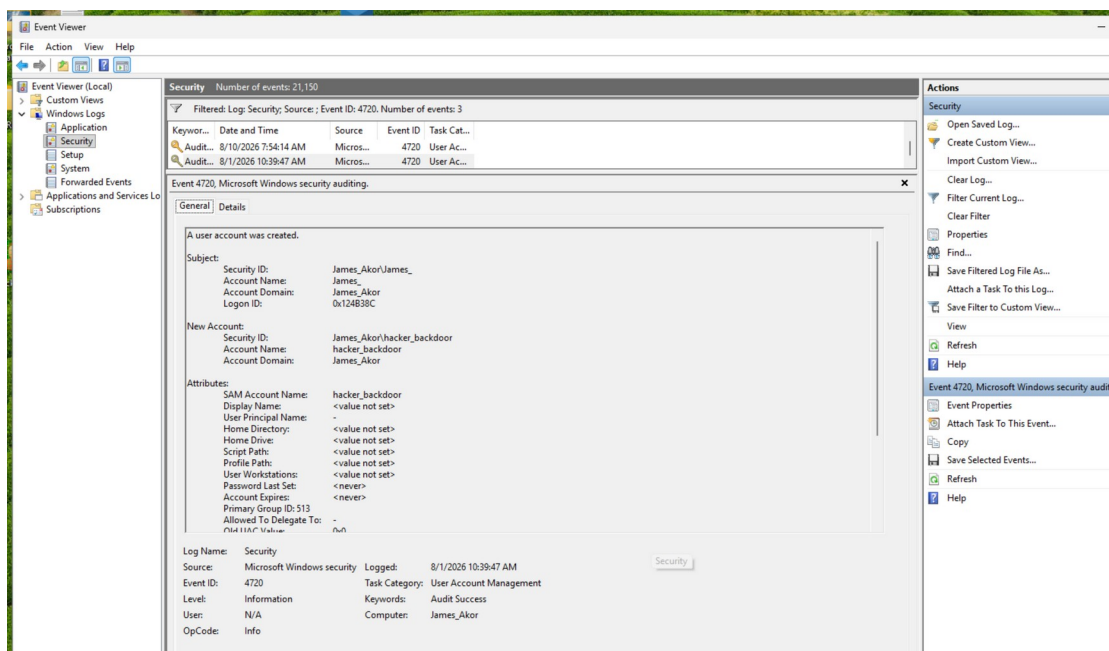


Figure B1 — Event 4720 — hacker_backdoor account creation.

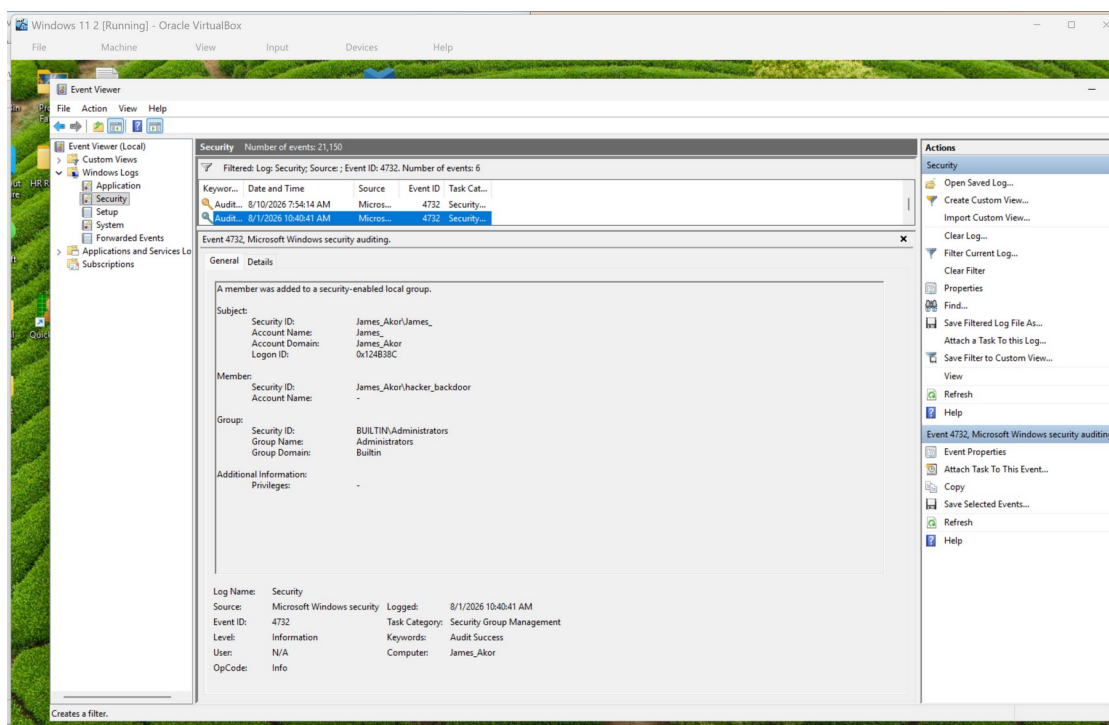


Figure B2 — Event 4732 — hacker_backdoor added to BUILTIN\Administrators.

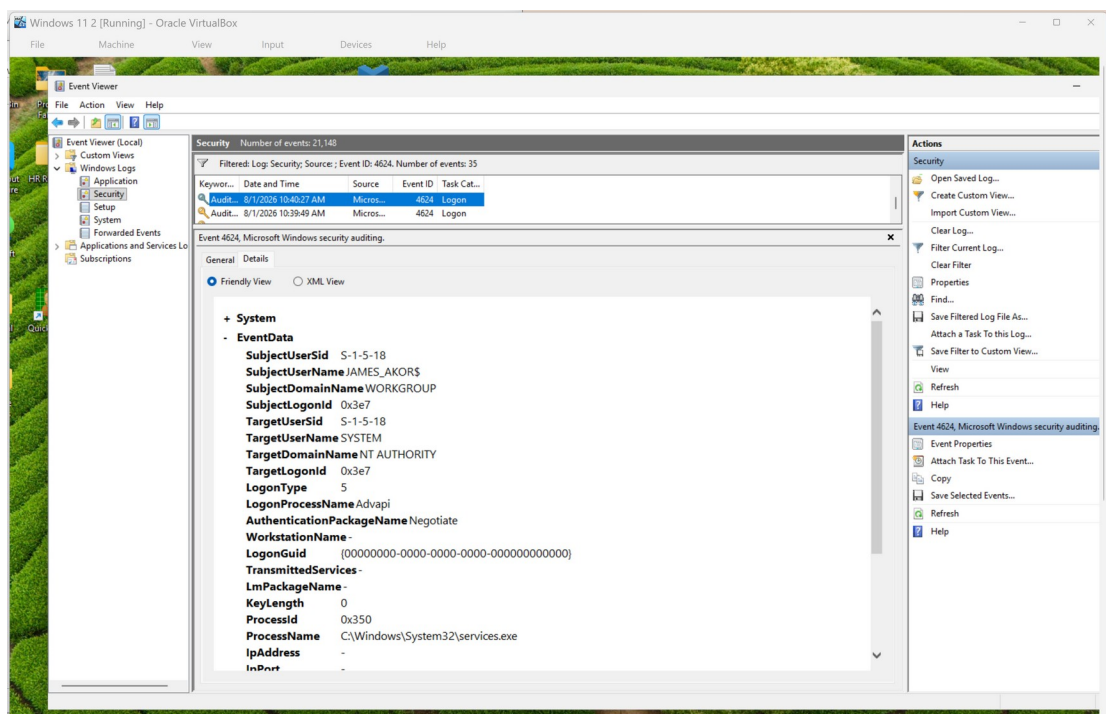


Figure B3 — Event 4624 — retained event is Logon Type 5.

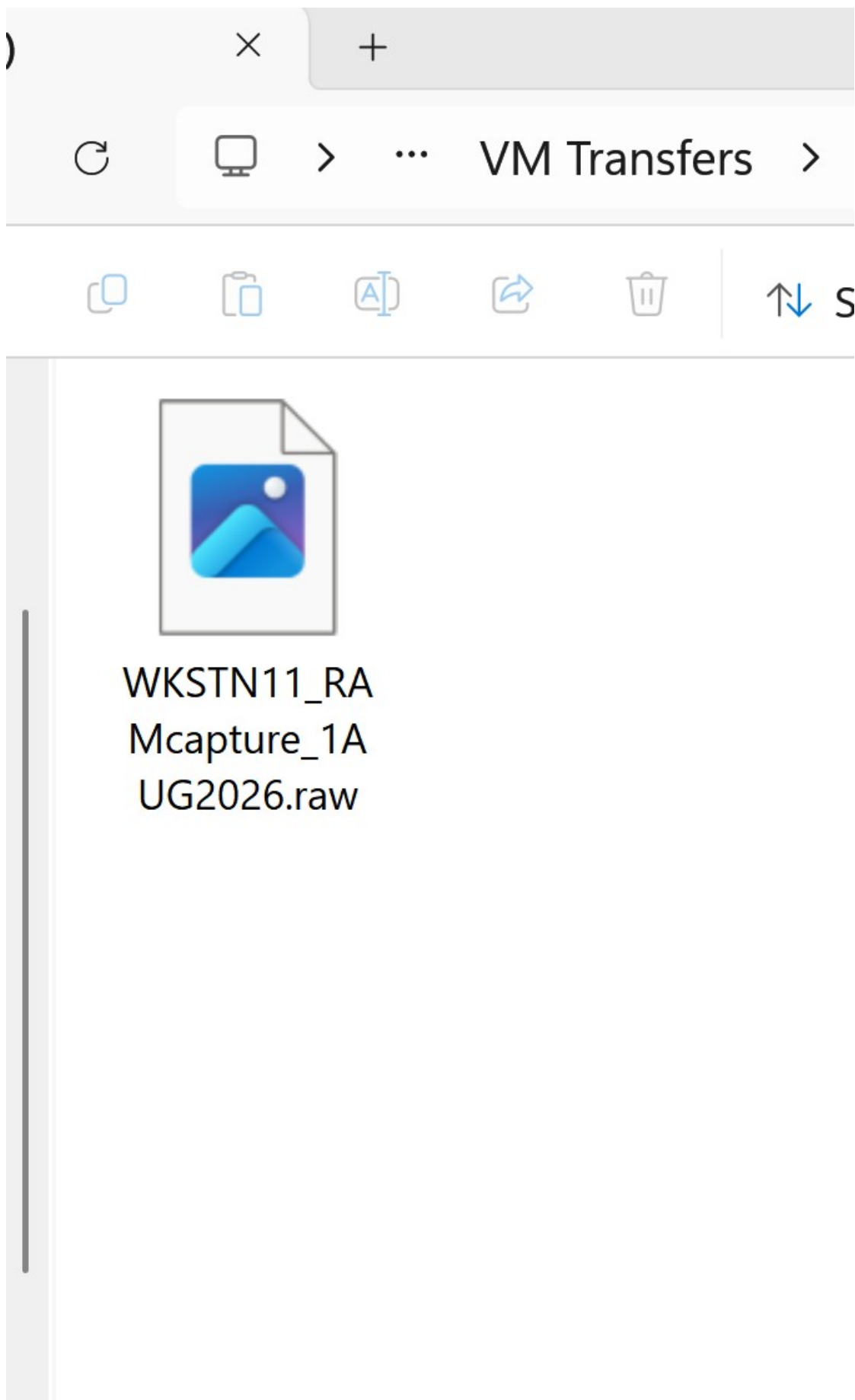


Figure B4 — Completed RAW memory capture.

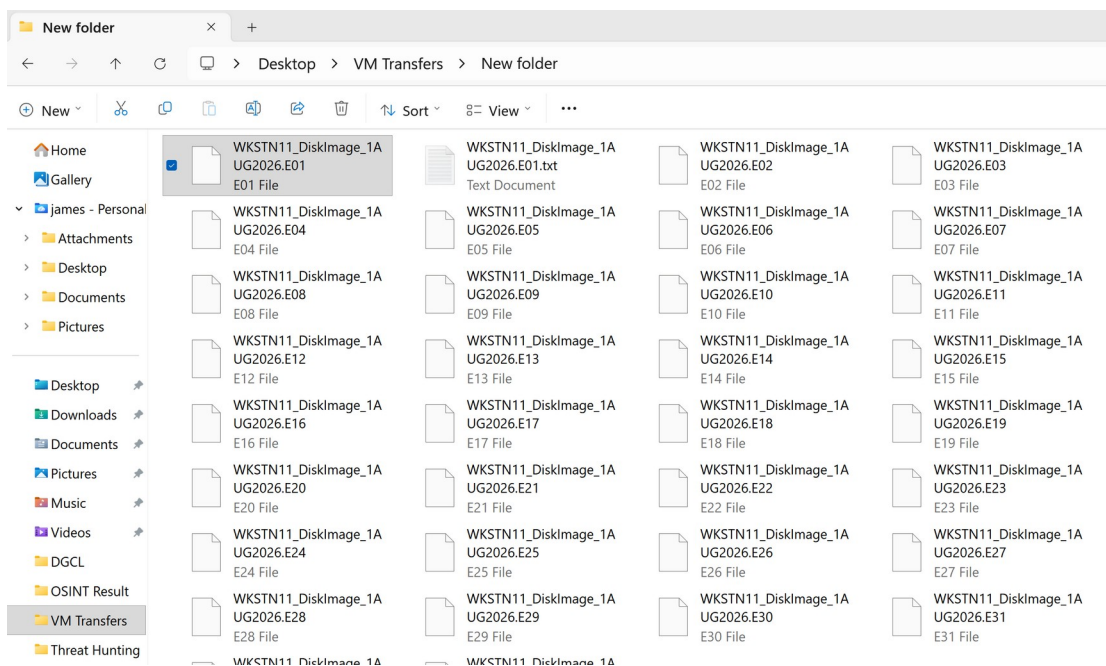


Figure B5 — Segmented E01 disk image in VM Transfers.

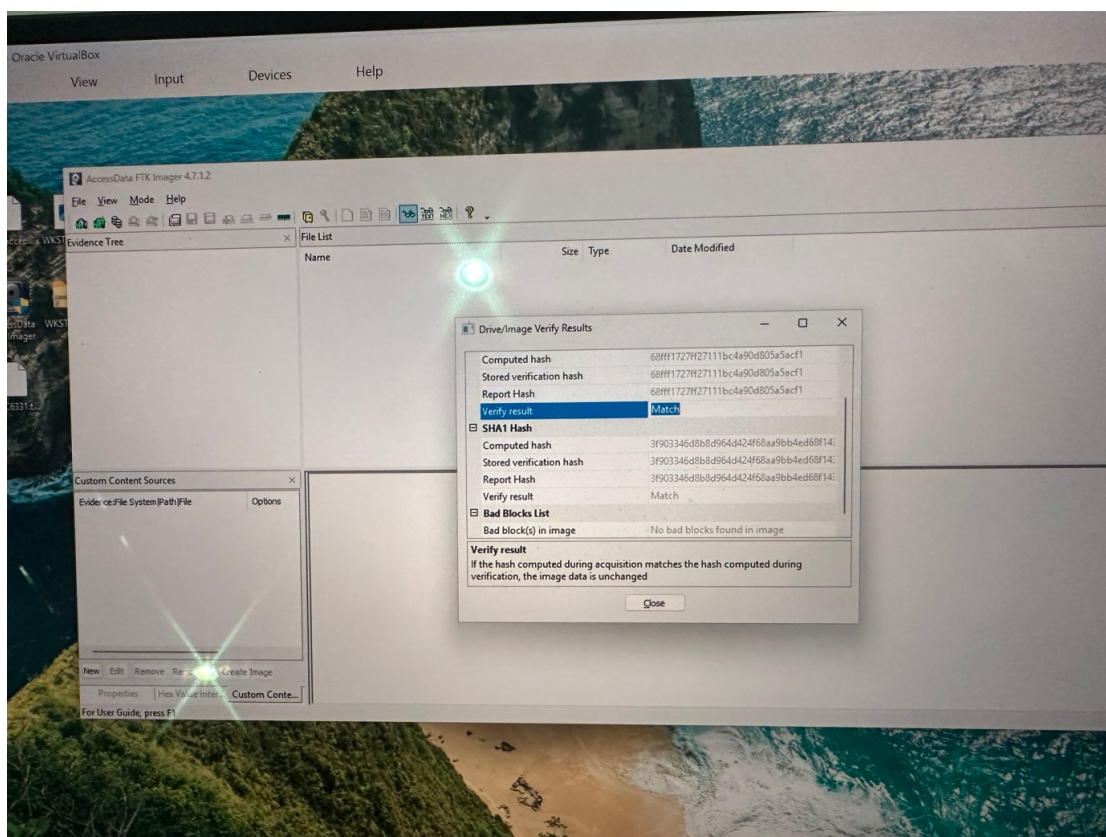
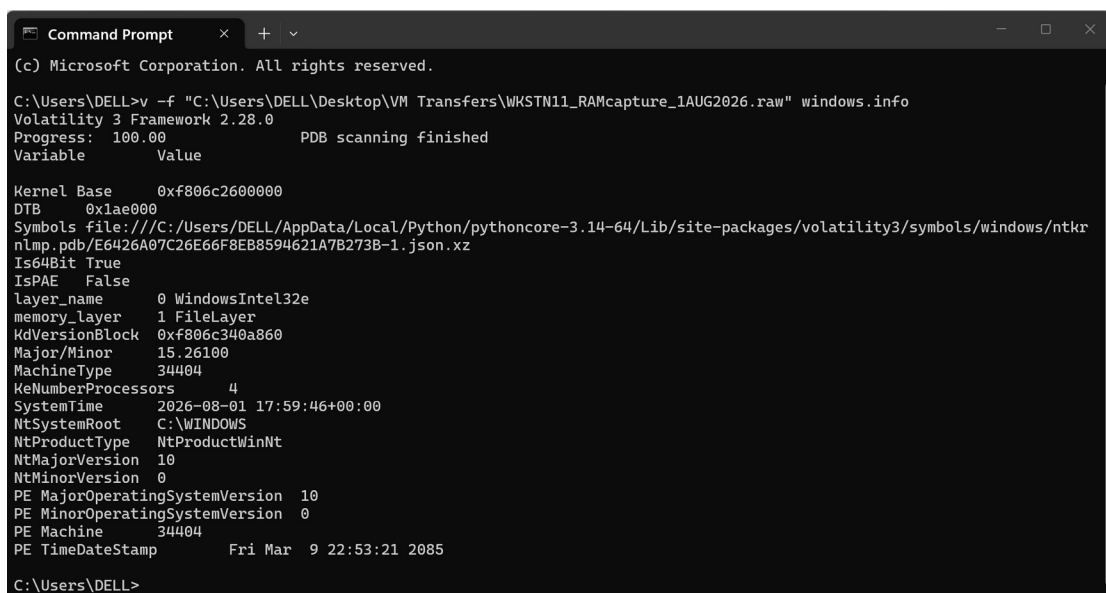


Figure B6 — FTK verification match and no bad blocks.



```

Command Prompt
(c) Microsoft Corporation. All rights reserved.

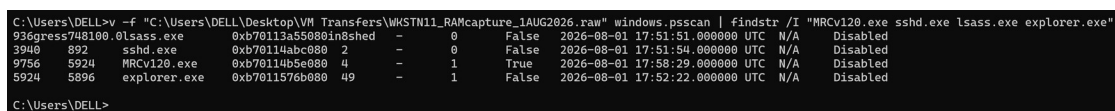
C:\Users\DELL>v -f "C:\Users\DELL\Desktop\VM Transfers\WKSTN11_RAMcapture_1AUG2026.raw" windows.info
Volatility 3 Framework 2.28.0
Progress: 100.00 PDB scanning finished
Variable Value

Kernel Base 0xf806c2600000
DTB 0x1ae000
Symbols file:///C:/Users/DELL/AppData/Local/Python/pythoncore-3.14-64/Lib/site-packages/volatility3/symbols/windows/ntkr
nlmp.pdb/E6426A07C26E66F8EB8594621A7B273B-1.json.xz
Is64Bit True
IsPAE False
layer_name 0 WindowsIntel32e
memory_layer 1 FileLayer
KdVersionBlock 0xf806c340a860
Major/Minor 15.26100
MachineType 34404
KeNumberProcessors 4
SystemTime 2026-08-01 17:59:46+00:00
NtSystemRoot C:\WINDOWS
NtProductType NtProductWinNt
NtMajorVersion 10
NtMinorVersion 0
PE MajorOperatingSystemVersion 10
PE MinorOperatingSystemVersion 0
PE Machine 34404
PE TimeDateStamp Fri Mar 9 22:53:21 2085

C:\Users\DELL>

```

Figure B7 — Volatility windows.info.



```

C:\Users\DELL>v -f "C:\Users\DELL\Desktop\VM Transfers\WKSTN11_RAMcapture_1AUG2026.raw" windows.psscan | findstr /I "MRCv120.exe sshd.exe lsass.exe explorer.exe"
936 gress748100.0lsass.exe 0xb70113a55080in8shed - 0 False 2026-08-01 17:51:51.000000 UTC N/A Disabled
3940 892 sshd.exe 0xb70114abc080 2 - 0 False 2026-08-01 17:51:54.000000 UTC N/A Disabled
9756 5924 MRCv120.exe 0xb70114b5e080 4 - 1 True 2026-08-01 17:58:29.000000 UTC N/A Disabled
5924 5896 explorer.exe 0xb7011576b080 49 - 1 False 2026-08-01 17:52:22.000000 UTC N/A Disabled

C:\Users\DELL>

```

Figure B8 — Volatility process analysis.

```

C:\Users\DELL>v -f "C:\Users\DELL\Desktop\VM Transfers\WKSTN11_RAMcapture_1AUG2026.raw" windows.netstat | findstr /I "sshd.exe 3389"
0xb7010e4f8488.0TCPv4 0.0.0.0 228 scan0.0.0.0n0shed LISTENING 3940 sshd.exe 2026-08-01 17:51:56.000000 UTC
0xb7010e4f8488 TCPv6 :: 22 :: 0 LISTENING 3940 sshd.exe 2026-08-01 17:51:56.000000 UTC
0xb7010e4f8cc8 TCPv4 0.0.0.0 22 0.0.0.0 0 LISTENING 3940 sshd.exe 2026-08-01 17:51:56.000000 UTC
0xb7011310d248 TCPv4 0.0.0.0 3389 0.0.0.0 0 LISTENING 1348 svchost.exe 2026-08-01 17:51:52.000000 UTC
0xb7011310d248 TCPv6 :: 3389 :: 0 LISTENING 1348 svchost.exe 2026-08-01 17:51:52.000000 UTC
0xb701139e8508 TCPv4 0.0.0.0 3389 0.0.0.0 0 LISTENING 1348 svchost.exe 2026-08-01 17:51:52.000000 UTC
0xb701140bd2b8 UDPv4 0.0.0.0 3389 * 0 1348 svchost.exe 2026-08-01 17:51:52.000000 UTC
0xb701140bd2b8 UDPv6 :: 3389 * 0 1348 svchost.exe 2026-08-01 17:51:52.000000 UTC
0xb701140bde18 UDPv4 0.0.0.0 3389 * 0 1348 svchost.exe 2026-08-01 17:51:52.000000 UTC
C:\Users\DELL>

```

Figure B9 — Volatility network listeners.

```

C:\Users\DELL>v -f "C:\Users\DELL\Desktop\VM Transfers\WKSTN11_RAMcapture_1AUG2026.raw" windows.malware.psxview | findstr /I "False"
0xb7011487b080.0audiodg.exe 3400scanFalsefinTrued False False 2026-08-01 18:03:37+00:00
0xb7010b697040 System 4 True False True False
0xb7010efef040 smss.exe 504 True True True False
0xb7010b69f040 MemCompression 2768 True True True False
0xb701155cd080 svchost.exe 5600 False True False False 2026-08-01 18:04:21+00:00
0xb7011648f080 WindowsTermina 3088 False True False False 2026-08-01 17:58:31+00:00
0xb7010b73e080 Registry 128 True True True False
0xb7011399f080 svchost.exe 5520 True True True False 2026-08-01 18:07:14+00:00
0xb70114a9f080 WmiPrvSE.exe 7544 False True False False 2026-08-01 18:03:54+00:00
0xb70115c80080 svchost.exe 5784 True True False False 2026-08-01 17:56:17+00:00
0xb70113c290c0 WerFault.exe 916 True True False False 2026-08-01 17:52:27+00:00
0xb7011490c080 SearchFilterHo 6716 False True False False 2026-08-01 17:54:48+00:00
0xb70113e52080 svchost.exe 1336 False True False False 2026-08-01 18:01:46+00:00
0xb70114281080 svchost.exe 8416 False True True False 2026-08-01 18:07:54+00:00
0xb70113306140 csrss.exe 676 True True True False
0xb70113a63140 csrss.exe 756 True True True False
0xb701166cf080 svchost.exe 10060 False True False False 2026-08-01 17:54:02+00:00
0xb7010b7b0080 svchost.exe 5700 False True False False 2026-08-01 17:59:14+00:00
0xb70115648080 userinit.exe 5896 True True False False 2026-08-01 17:52:54+00:00
0xb70114e2d080 MoUsoCoreWorke 3652 False True True False 2026-08-01 18:07:56+00:00
0xb70114c73080 smartscreen.ex 8644 False True True False 2026-08-01 18:07:44+00:00
0xb701163f3080 sppsvc.exe 10096 False True False False 2026-08-01 17:54:37+00:00
C:\Users\DELL>

```

Figure B10 — PSXView false-result review.

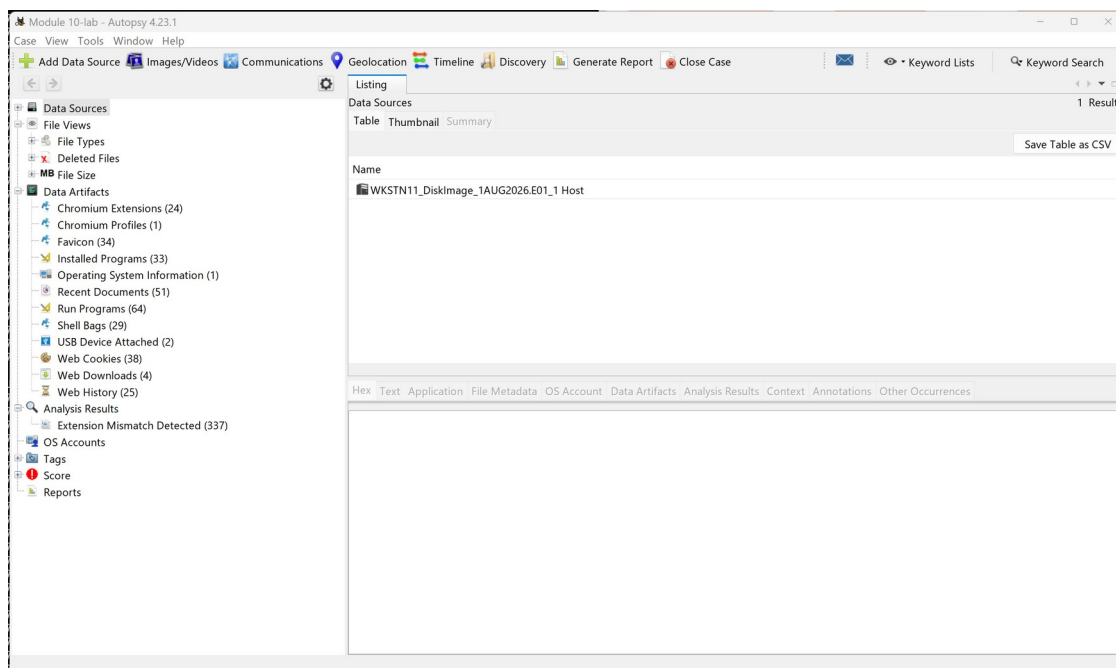


Figure B11 — Autopsy case and E01 evidence source.

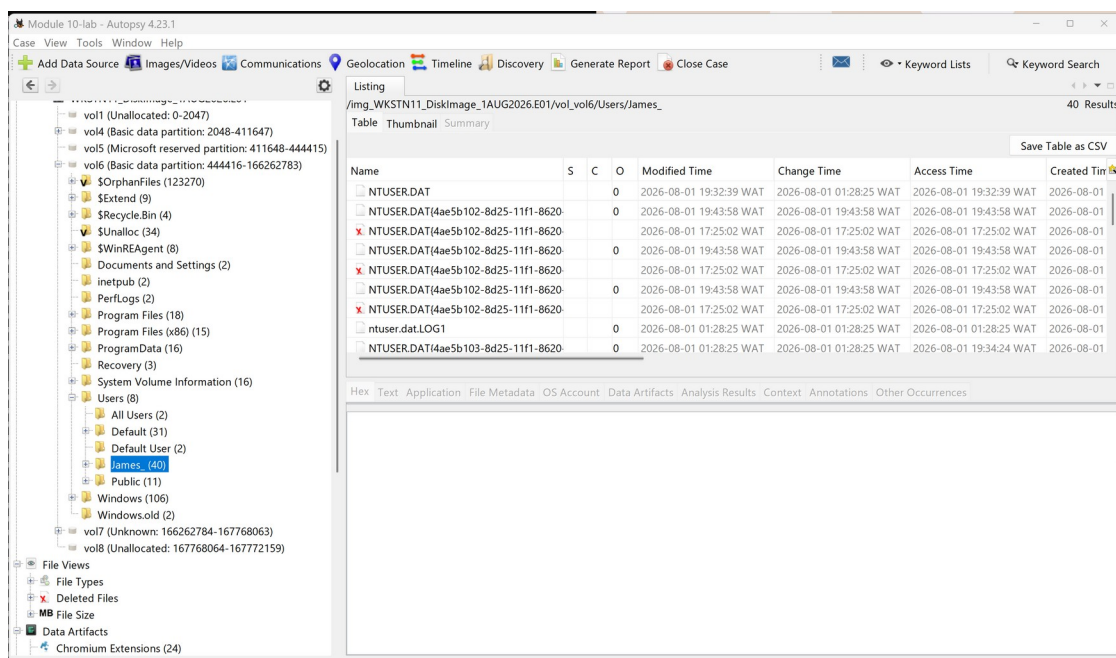


Figure B12 — Main Windows partition / James_ profile.

| Name | S | C | O | Modified Time | Change Time | Access Time | Created Time | Size |
|---|---|---|---|-------------------------|-------------------------|-------------------------|-------------------------|------------|
| content.bin | | | | 2026-08-01 02:33:13 WAT | 2026-08-01 02:33:13 WAT | 2026-08-01 02:33:13 WAT | 2026-08-01 02:33:13 WAT | 3462449862 |
| content.bin | | | | 2026-08-01 17:49:05 WAT | 2026-08-01 17:49:05 WAT | 2026-08-01 17:49:05 WAT | 2026-08-01 17:49:05 WAT | 757341203 |
| content.bin | | | | 2026-08-01 18:19:00 WAT | 2026-08-01 18:19:00 WAT | 2026-08-01 18:19:00 WAT | 2026-08-01 18:19:00 WAT | 309518653 |
| content.bin | | | | 2026-08-01 18:29:09 WAT | 2026-08-01 18:29:09 WAT | 2026-08-01 18:29:09 WAT | 2026-08-01 18:29:09 WAT | 298707463 |
| content.bin | | | | 2026-08-01 18:18:08 WAT | 2026-08-01 18:18:08 WAT | 2026-08-01 18:18:08 WAT | 2026-08-01 18:18:08 WAT | 90981314 |
| content.bin | | | | 2026-07-31 14:24:10 WAT | 2026-07-31 14:24:10 WAT | 2026-07-31 14:24:10 WAT | 2026-07-31 14:24:10 WAT | 45165195 |
| vmfirmwarehcl.dll | | | | 2026-08-01 18:50:57 WAT | 2026-08-01 18:51:13 WAT | 2026-08-01 18:50:57 WAT | 2026-08-01 18:51:13 WAT | 36177400 |
| content.bin | | | | 2026-07-31 14:38:39 WAT | 2026-07-31 14:38:39 WAT | 2026-07-31 14:38:39 WAT | 2026-07-31 14:38:39 WAT | 29981141 |
| WindowsOobeAppHost.AOT.dll | | | | 2026-08-01 18:29:10 WAT | 2026-08-01 18:52:03 WAT | 2026-08-01 18:52:00 WAT | 2026-08-01 18:52:00 WAT | 26561536 |
| content.bin | | | | 2026-08-01 18:59:10 WAT | 2026-08-01 18:59:10 WAT | 2026-08-01 18:59:10 WAT | 2026-08-01 18:59:10 WAT | 25513625 |
| FaceRecognitionEngineAdapterResource | | | | 2026-08-01 18:25:19 WAT | 2026-08-01 18:52:41 WAT | 2026-08-01 18:52:41 WAT | 2026-08-01 18:52:41 WAT | 25029112 |
| content.bin | | | | 2026-08-01 18:26:27 WAT | 2026-08-01 18:26:27 WAT | 2026-08-01 18:26:27 WAT | 2026-08-01 18:26:27 WAT | 22584411 |
| content.bin | | | | 2026-08-01 18:19:03 WAT | 2026-08-01 18:19:03 WAT | 2026-08-01 18:19:03 WAT | 2026-08-01 18:19:03 WAT | 22305047 |
| facerecognitionengineadapterresources | | | | 2026-08-01 18:25:20 WAT | 2026-08-01 18:52:26 WAT | 2026-08-01 18:52:26 WAT | 2026-08-01 18:52:26 WAT | 21355800 |
| mpengine.dll | | | | 2026-08-01 17:46:25 WAT | 2026-08-01 17:46:43 WAT | 2026-08-01 18:32:18 WAT | 2026-08-01 17:46:43 WAT | 19187008 |
| FaceDetectorResources.dll | | | | 2026-08-01 18:25:19 WAT | 2026-08-01 18:52:40 WAT | 2026-08-01 18:52:40 WAT | 2026-08-01 18:52:40 WAT | 19103824 |
| data_3 | | | | 2026-08-01 19:58:37 WAT | 2026-08-01 19:58:37 WAT | 2026-08-01 19:58:37 WAT | 2026-08-01 01:13:00 WAT | 16785408 |
| Microsoft.luixaml.dll | | | | 2026-08-01 18:29:18 WAT | 2026-08-01 18:52:46 WAT | 2026-08-01 18:52:46 WAT | 2026-08-01 18:52:46 WAT | 15079936 |
| onmxruntime.dll | | | | 2026-08-01 18:29:16 WAT | 2026-08-01 18:53:04 WAT | 2026-08-01 18:53:04 WAT | 2026-08-01 18:53:04 WAT | 14180904 |
| facerecognitionengineadapterresources | | | | 2026-08-01 18:25:19 WAT | 2026-08-01 18:52:34 WAT | 2026-08-01 18:52:34 WAT | 2026-08-01 18:52:34 WAT | 13653752 |
| vmfirmwarehcl.dll | | | | 2026-08-01 18:50:57 WAT | 2026-08-01 18:51:29 WAT | 2026-08-01 18:54:50 WAT | 2026-08-01 18:51:29 WAT | 12129138 |
| TextInput.dll | | | | 2026-08-01 18:29:17 WAT | 2026-08-01 18:52:56 WAT | 2026-08-01 18:52:56 WAT | 2026-08-01 18:52:56 WAT | 11123712 |
| DynamicLottie.dll | | | | 2026-08-01 18:29:17 WAT | 2026-08-01 18:52:04 WAT | 2026-08-01 18:52:00 WAT | 2026-08-01 18:52:00 WAT | 10801664 |
| WXHwIntAdapter.dll | | | | 2026-08-01 18:29:10 WAT | 2026-08-01 18:49:50 WAT | 2026-08-01 18:48:11 WAT | 2026-08-01 18:48:11 WAT | 10574363 |
| onmxruntime.dll | | | | 2026-08-01 18:25:46 WAT | 2026-08-01 18:51:54 WAT | 2026-08-01 18:51:48 WAT | 2026-08-01 18:51:48 WAT | 10572960 |
| Microsoft.CognitiveServices.Speech.External | | | | 2026-08-01 18:29:16 WAT | 2026-08-01 18:53:07 WAT | 2026-08-01 18:53:07 WAT | 2026-08-01 18:53:07 WAT | 10517536 |

Figure B13 — Deleted-file results sorted by size.

| Source Name | S | C | O | Program Name | Username | Date/Time | Comment |
|-------------|---|---|---|---|--------------|-------------------------|--------------------------------|
| SYSTEM | | | | Windows\System32\cmd.exe | | 2026-08-01 18:50:29 WAT | Recently executed according to |
| SYSTEM | | | | Windows\System32\WindowsPowerShell\cmd.exe | | 2026-08-01 19:26:32 WAT | Recently Executed according to |
| SYSTEM | | | | Microsoft.SecHealthUI\Bwekyb3d8bbwe | | 2026-08-01 19:26:41 WAT | Recently Executed according to |
| SYSTEM | | | | Windows\System32\Taskmgr.exe | | 2026-08-01 02:42:45 WAT | Recently Executed according to |
| SYSTEM | | | | Microsoft.Windows.ShellExperienceHost\cmd.exe | | 2026-08-01 19:32:37 WAT | Recently Executed according to |
| SYSTEM | | | | Microsoft.XboxGamingOverlay\Bwekyb3d8bbwe | | 2026-08-01 17:52:32 WAT | Recently Executed according to |
| SYSTEM | | | | Windows\System32\OpenWith.exe | | 2026-08-01 18:58:14 WAT | Recently Executed according to |
| SYSTEM | | | | Microsoft.WindowsNotepad\Bwekyb3d8bbwe | | 2026-08-01 19:26:47 WAT | Recently Executed according to |
| SYSTEM | | | | Windows\System32\DataExchangeHost.exe | | 2026-08-01 19:10:39 WAT | Recently Executed according to |
| SYSTEM | | | | Program Files (x86)\Microsoft\Edge\Appl... | | 2026-08-01 19:46:28 WAT | Recently Executed according to |
| SYSTEM | | | | Microsoft.LockApp_cw5n1h2zyewy | | 2026-08-01 20:06:55 WAT | Recently Executed according to |
| SYSTEM | | | | Program Files\Oracle\VirtualBox Guest Ad... | | 2026-08-01 19:27:53 WAT | Recently Executed according to |
| SYSTEM | | | | Windows\System32\cmd.exe | | 2026-08-01 12:11:49 WAT | Recently Executed according to |
| SYSTEM | | | | Program Files (x86)\Quick Stego\quickste... | | 2026-08-01 12:18:10 WAT | Recently Executed according to |
| SYSTEM | | | | Users\James\Desktop\MRCv120.exe | | 2026-08-01 18:58:33 WAT | Recently Executed according to |
| SYSTEM | | | | Windows\System32\cmd.exe | S-1-5-90-0-1 | 2026-08-02 03:43:40 WAT | Recently Executed according to |

| Type | Value | Source(s) |
|------------------|---|-------------------------|
| Program Name | Users\James\Desktop\MRCv120.exe | Windows Registry Analyz |
| Username | | Windows Registry Analyz |
| Date/Time | 2026-08-01 18:58:33 WAT | Windows Registry Analyz |
| Comment | Recently Executed according to Background Activity Moderator (BAM) | Windows Registry Analyz |
| Source File Path | /img_WKSTN11_DiskImage_1AUG2026.E01/vol6/Windows/System32/config/SYSTEM | |
| Artifact ID | -9223372036854775244 | |

Figure B14 — BAM execution artefact for MRCv120.exe.

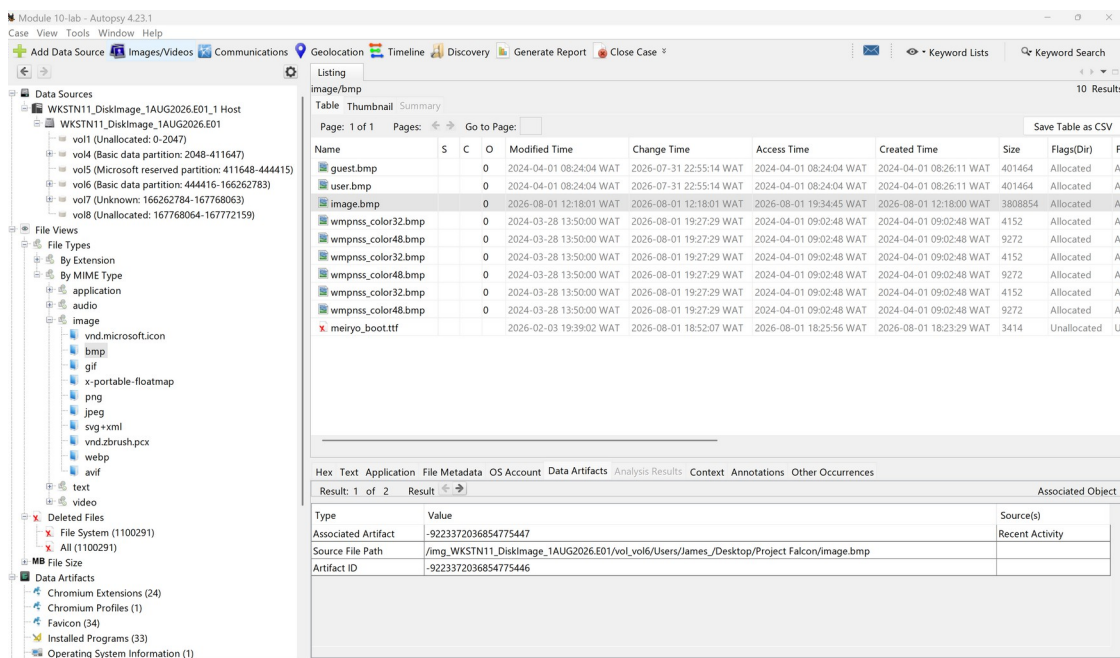


Figure B15 — image.bmp located in Project Falcon.

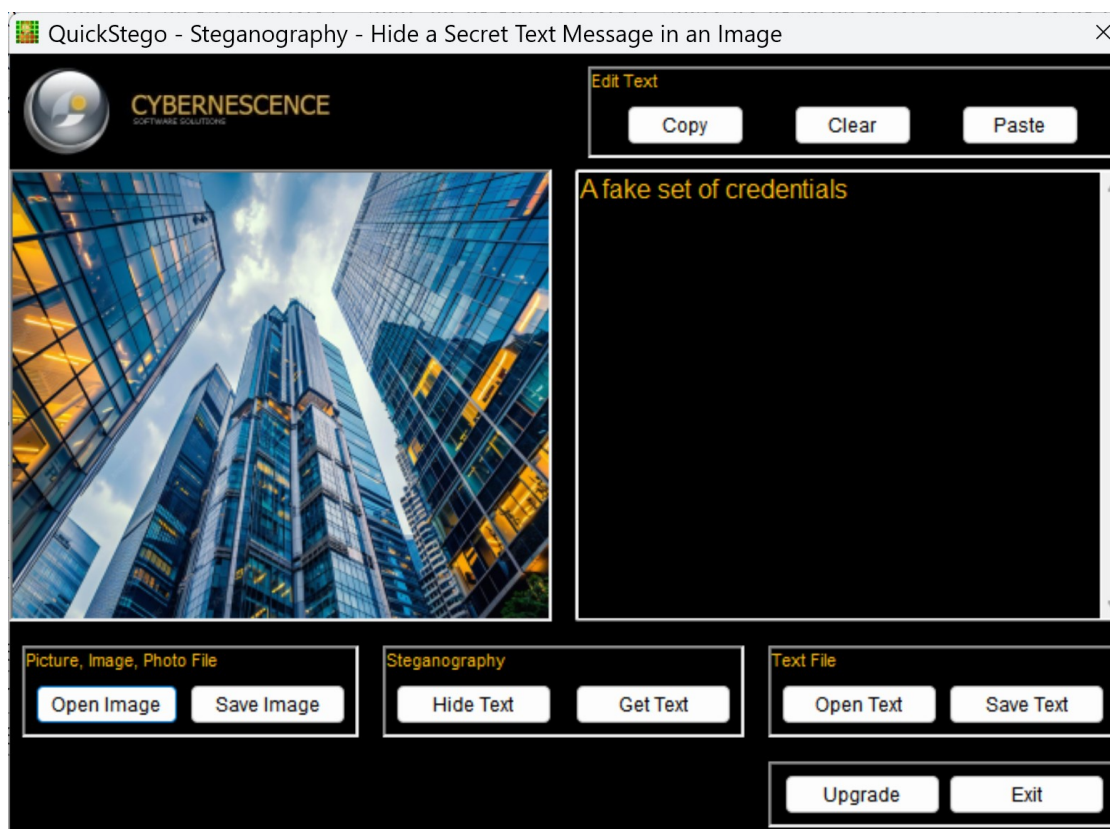
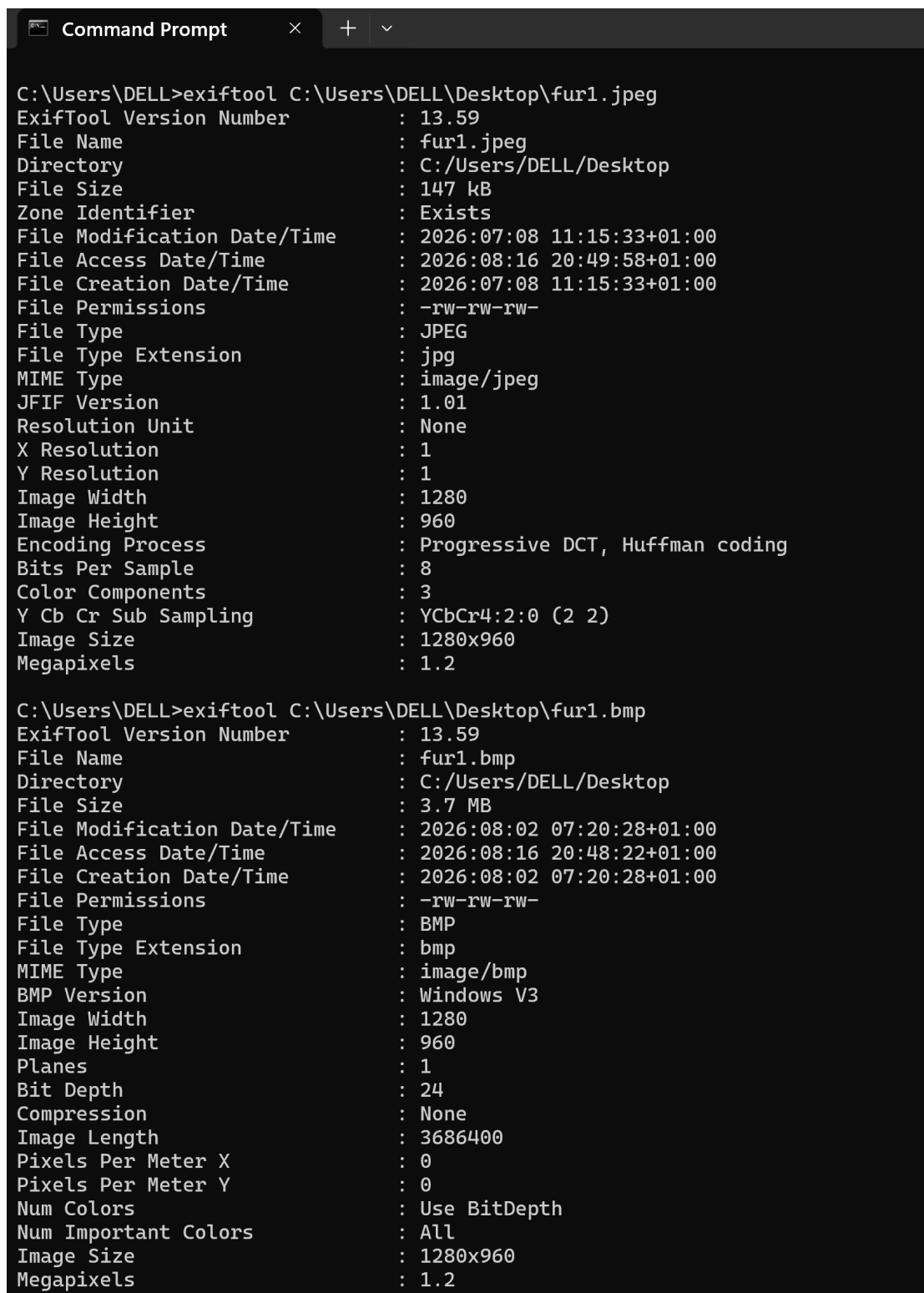


Figure B16 — QuickStego recovered hidden message.



```
Command Prompt
C:\Users\DELL>exiftool C:\Users\DELL\Desktop\fur1.jpeg
ExifTool Version Number      : 13.59
File Name                    : fur1.jpeg
Directory                   : C:/Users/DELL/Desktop
File Size                    : 147 kB
Zone Identifier              : Exists
File Modification Date/Time   : 2026:07:08 11:15:33+01:00
File Access Date/Time        : 2026:08:16 20:49:58+01:00
File Creation Date/Time      : 2026:07:08 11:15:33+01:00
File Permissions             : -rw-rw-rw-
File Type                    : JPEG
File Type Extension          : jpg
MIME Type                    : image/jpeg
JFIF Version                 : 1.01
Resolution Unit              : None
X Resolution                  : 1
Y Resolution                  : 1
Image Width                  : 1280
Image Height                  : 960
Encoding Process              : Progressive DCT, Huffman coding
Bits Per Sample              : 8
Color Components              : 3
Y Cb Cr Sub Sampling         : YCbCr4:2:0 (2 2)
Image Size                   : 1280x960
Megapixels                   : 1.2

C:\Users\DELL>exiftool C:\Users\DELL\Desktop\fur1.bmp
ExifTool Version Number      : 13.59
File Name                    : fur1.bmp
Directory                   : C:/Users/DELL/Desktop
File Size                    : 3.7 MB
File Modification Date/Time   : 2026:08:02 07:20:28+01:00
File Access Date/Time        : 2026:08:16 20:48:22+01:00
File Creation Date/Time      : 2026:08:02 07:20:28+01:00
File Permissions             : -rw-rw-rw-
File Type                    : BMP
File Type Extension          : bmp
MIME Type                    : image/bmp
BMP Version                  : Windows V3
Image Width                  : 1280
Image Height                  : 960
Planes                       : 1
Bit Depth                    : 24
Compression                  : None
Image Length                  : 3686400
Pixels Per Meter X           : 0
Pixels Per Meter Y           : 0
Num Colors                    : Use BitDepth
Num Important Colors          : All
Image Size                   : 1280x960
Megapixels                   : 1.2
```

Figure B17 — ExifTool JPEG/BMP comparison.

Appendix C — Core Reproducibility Commands⁴¹

The commands below capture the principal actions used in the lab. Credentials are intentionally represented as a placeholder rather than published as part of the portfolio.

| Action | Command |
|---------------------------|--|
| Account creation | <code>net user hacker_backdoor <lab-password> /add</code> |
| Administrator membership | <code>net localgroup Administrators hacker_backdoor /add</code> |
| RDP listener check | <code>netstat -ano findstr :3389</code> |
| RAM SHA-256 | <code>certutil -hashfile WKSTN11_RAMcapture_1AUG2026.raw SHA256</code> |
| Volatility validation | <code>v -h</code> |
| Memory profile validation | <code>v -f "C:\PATH\TO\WKSTN11_RAMcapture_1AUG2026.raw" windows.info</code> |
| Process scan | <code>v -f "C:\PATH\TO\WKSTN11_RAMcapture_1AUG2026.raw" windows.psscan</code> |
| Network scan | <code>v -f "C:\PATH\TO\WKSTN11_RAMcapture_1AUG2026.raw" windows.netscan</code> |
| PSXView | <code>v -f "C:\PATH\TO\WKSTN11_RAMcapture_1AUG2026.raw" windows.malware.psxview</code> |
| ExifTool JPEG | <code>exiftool C:\Users\DELL\Desktop\fur1.jpeg</code> |
| ExifTool BMP | <code>exiftool C:\Users\DELL\Desktop\fur1.bmp</code> |

Appendix D — Evidence Limitations & Quality Notes⁴¹

- The retained RAM-hash screenshot shows the certutil command but not the resulting SHA-256 digest; the value is therefore not transcribed.
- The retained 4624 screenshot shows Logon Type 5. It should not be used as proof of an RDP Logon Type 10 event.
- Volatility shows TCP/3389 listening, which demonstrates service availability but does not by itself prove that a remote session occurred.
- Autopsy reported a very large deleted-file result set. The retained screenshots are dominated by Windows/application artefacts, so no user-document deletion claim is made.
- PSXView discrepancies are treated as leads requiring corroboration, not as standalone proof of malware or rootkit activity.
- The steganography message is synthetic (“A fake set of credentials”) and exists solely for the controlled laboratory demonstration.

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